

EXPLAINABLE ARTIFICIAL INTELLIGENCE

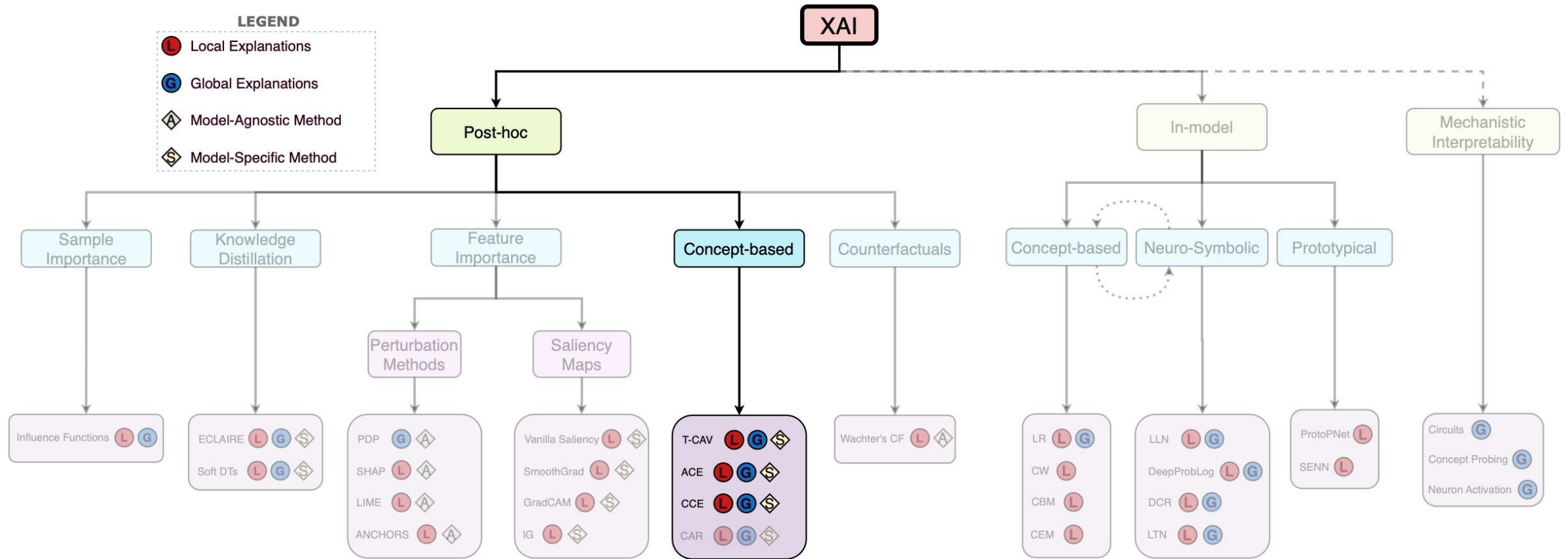
L193 – Lecture 4 – Lent 2025



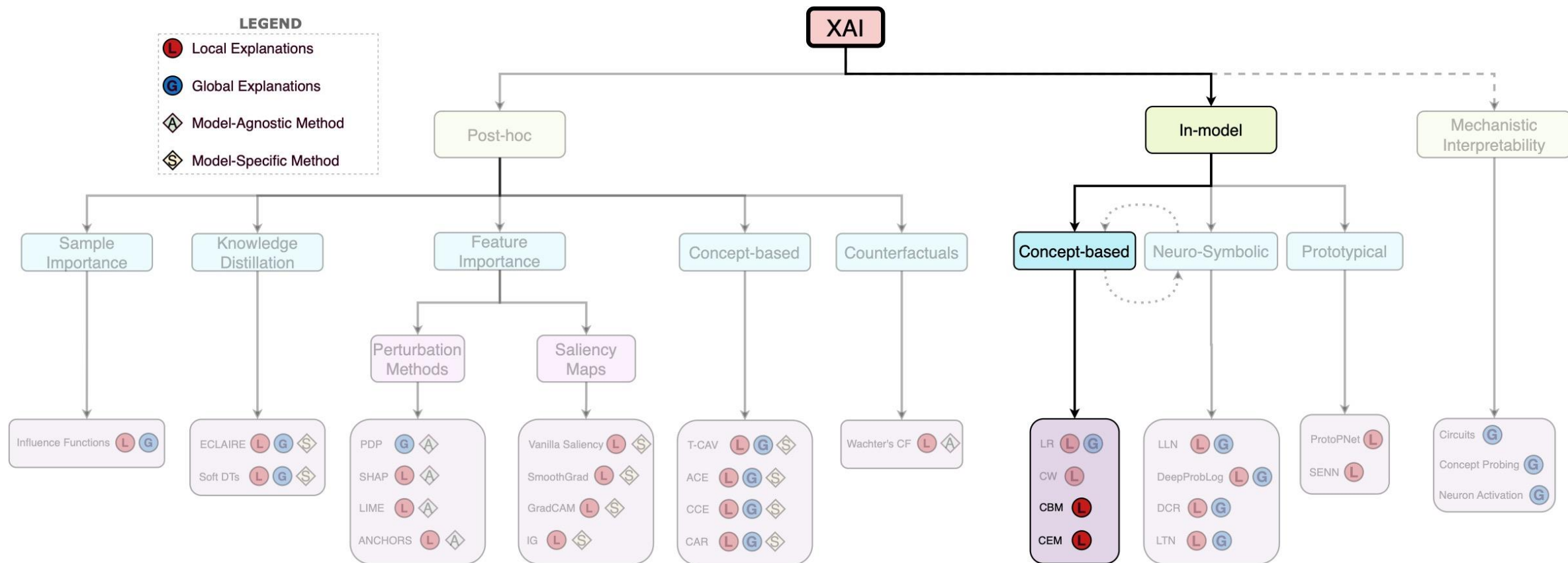
UNIVERSITY OF
CAMBRIDGE



WHERE WE LEFT OFF LAST TIME



TODAY: IN-MODEL EXPLAINABILITY



OUR XAI STORY SO FAR

We have focused on exploring **post-hoc** XAI methods



Taken from memecenter.com

OUR XAI STORY SO FAR

Post-hoc methods have a clear set of **important limitations**:

1. They may fail to properly explain a model → potentially **doubling the source of error!**



OUR XAI STORY SO FAR

Post-hoc methods have a clear set of **important limitations**:

1. They may fail to properly explain a model → potentially **doubling the source of error!**



In fact, **these methods often disagree** with each other (Krishna et al., [1])!



OUR XAI STORY SO FAR

Post-hoc methods have a clear set of **important limitations**:

2. They are **unable to capture causal relationships** between input features, concepts, and output labels

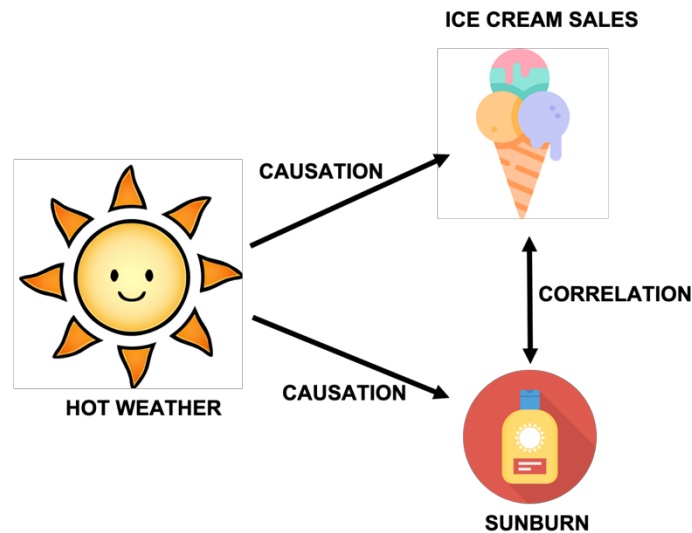


Image taken from <https://www.royriachi.com/2019/02/correlational-and-causal-relationships.html>

OUR XAI STORY SO FAR

Post-hoc methods have a clear set of **important limitations**:

3. Explanations are prone to **confirmation bias** (Bertrand et al., [1])!

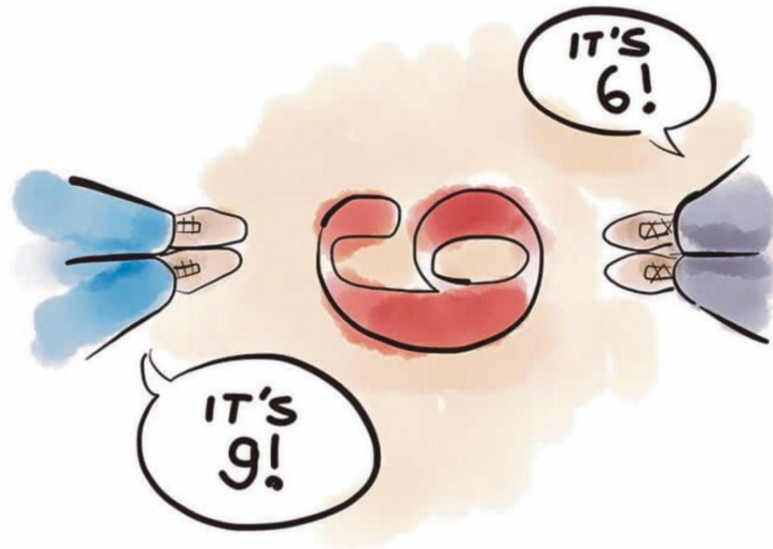


Image taken from "Confirmation Bias and the new Malaysia" by Datuk Steven Wong (New Straits Times)



SO, WHO WE'RE GOING TO CALL?

In-model Explainable Neural Architectures!

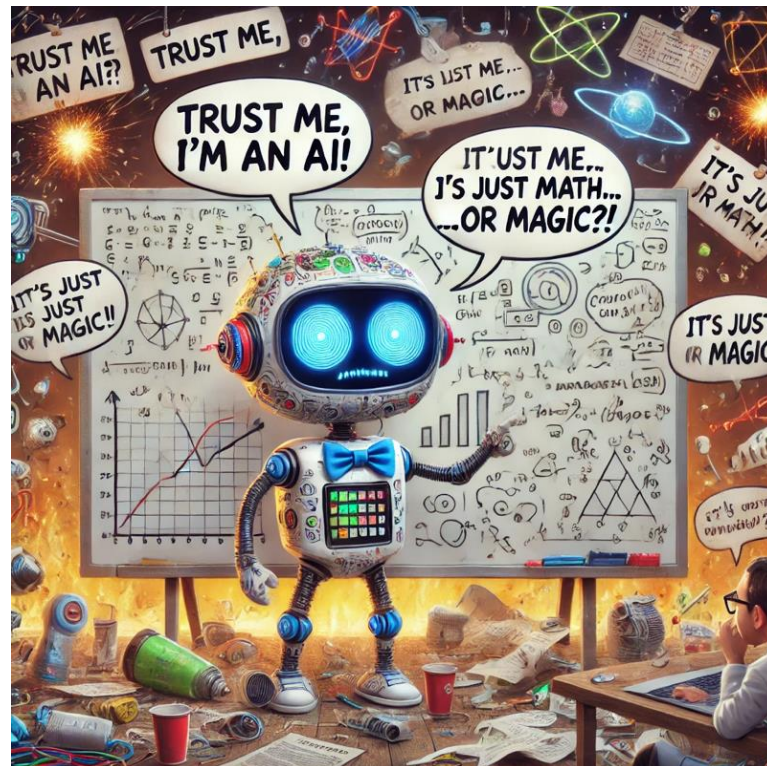


IN-MODEL EXPLAINABLE MODELS



GOING IN-MODEL

Rather than explaining an already trained model, **let the model explain itself!**



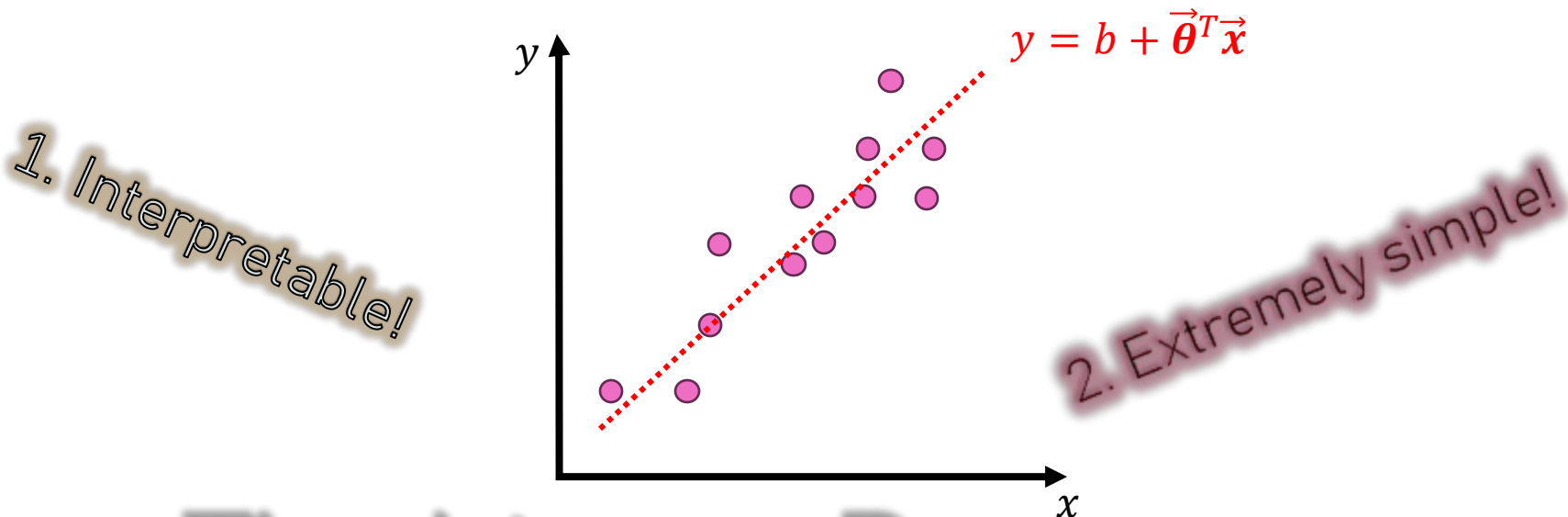
GOING IN-MODEL

There are a few main ways to achieve this:

GOING IN-MODEL

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1. **Linearity** (e.g., logistic regression)



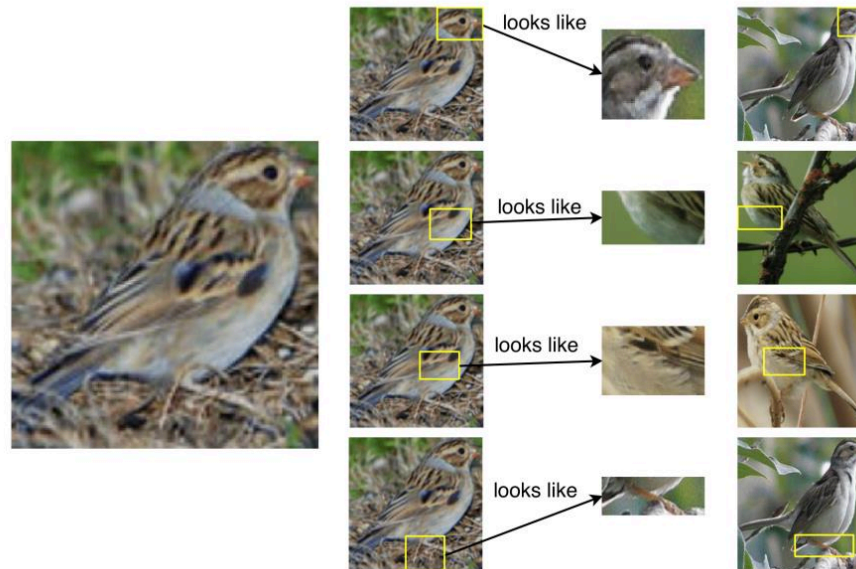
The Linear Regressor

3. Widely loved, understood, and revered!

GOING IN-MODEL

There are a few main ways to achieve this:

1. **Linearity** (e.g., logistic regression)
2. **Prototypical** explanations (e.g., "this looks like that")



GOING IN-MODEL

There are a few main ways to achieve this:

1. **Linearity** (e.g., logistic regression)
2. **Prototypical** explanations (e.g., "this looks like that")
3. And... **concepts!**

Today, we will **focus on concept-based interpretable architectures** but we will discuss **other alternatives in future lectures!**

CONCEPT BOTTLENECK MODELS (CBMS)

KOH ET AL. (ICML 2020)

Almost all concept-based interpretable architectures **can be framed** in terms of a **Concept Bottleneck Model** [1], or a CBM for short!

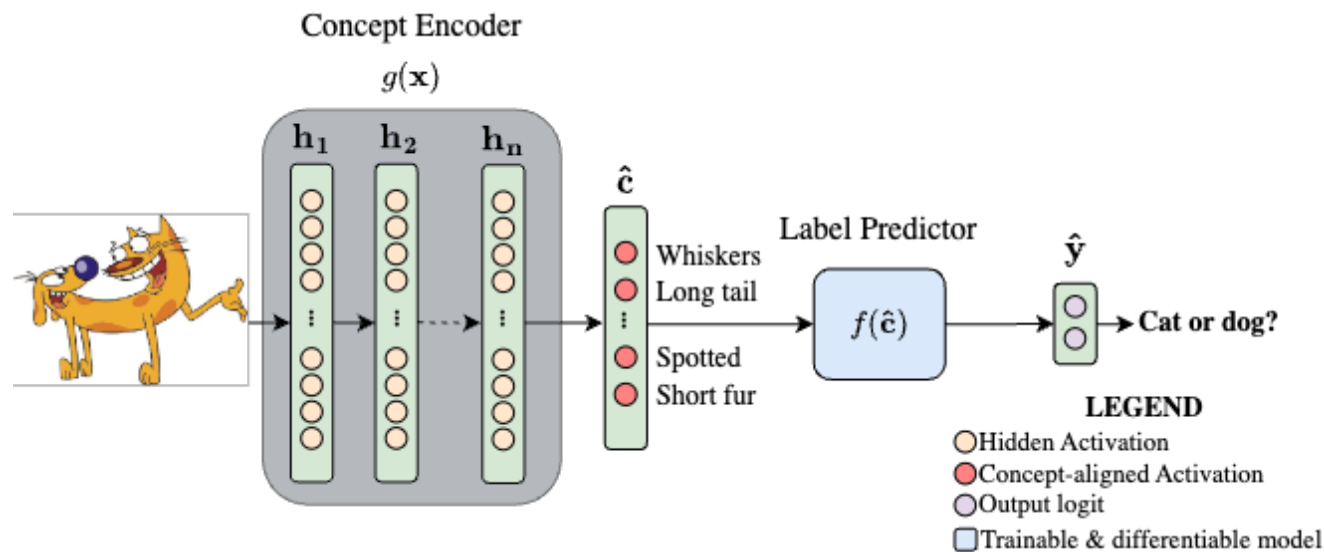


CONCEPT BOTTLENECK MODELS (CBMS)

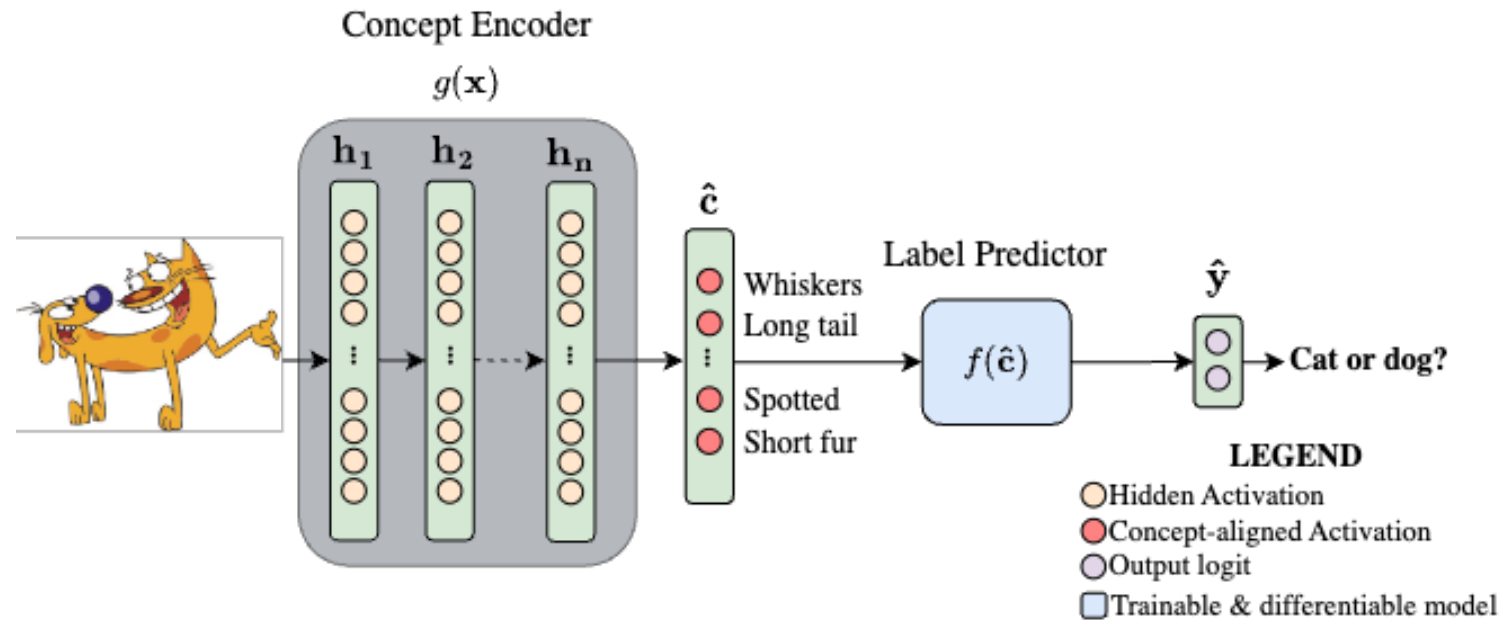
KOH ET AL. (ICML 2020)

CBMs force some of their latent spaces to be **aligned to known concepts** by composing two functions:

1. The first part will predict **concept activations from the input features** (concept encoder $g(\mathbf{x}) = \hat{\mathbf{c}}$)
2. The second part will **predict a task label from the predicted concepts** (label predictor $f(\hat{\mathbf{c}}) = \hat{y}$)



UNDERLYING ASSUMPTIONS



Assumptions for Concept Bottleneck Models (CBMs):

1. Each sample is annotated with a **task label** $y \in \{0, 1, \dots, L - 1\}$
2. Each sample is annotated with a vector $\mathbf{c} \in \{0, 1\}^k$ of k **binary concepts**

TRAINING A CBM

How would you **train** such a model **given a concept-annotated dataset** $\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{c}^{(i)}, y^{(i)})\}_{i=1}^N$?

(1) Independently

$$\begin{cases} \mathbb{E}_{(\mathbf{x}, \mathbf{c}) \sim \mathcal{D}} [\text{BCE}(g(\mathbf{x}), \mathbf{c})] \\ \mathbb{E}_{(\mathbf{c}, y) \sim \mathcal{D}} [\text{CE}(f(\mathbf{c}), y)] \end{cases}$$

(2) Sequentially

(a) $\mathbb{E}_{(\mathbf{x}, \mathbf{c}) \sim \mathcal{D}} [\text{BCE}(g(\mathbf{x}), \mathbf{c})]$

Freeze g

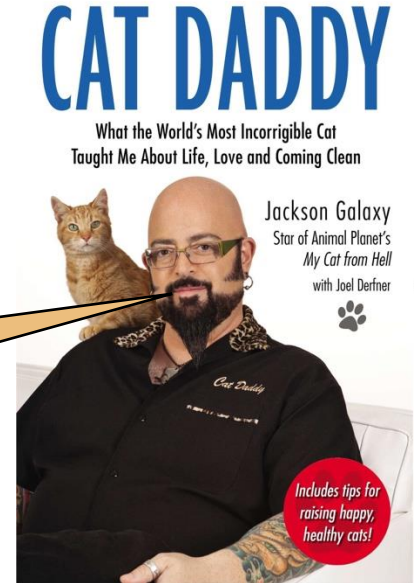
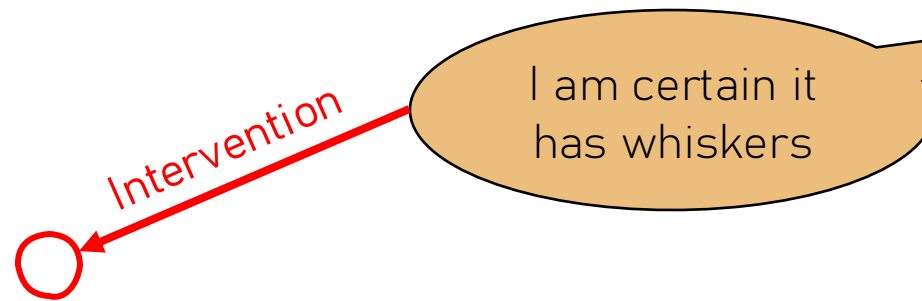
(b) $\mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [\text{CE}(f(g(\mathbf{x})), y)]$

(3) Jointly

$$\mathbb{E}_{(\mathbf{x}, \mathbf{c}, y) \sim \mathcal{D}} [\text{CE}(f(g(\mathbf{x})), y) + \lambda \cdot \text{BCE}(g(\mathbf{x}), \mathbf{c})]$$

Question: what does λ control for in the joint training?

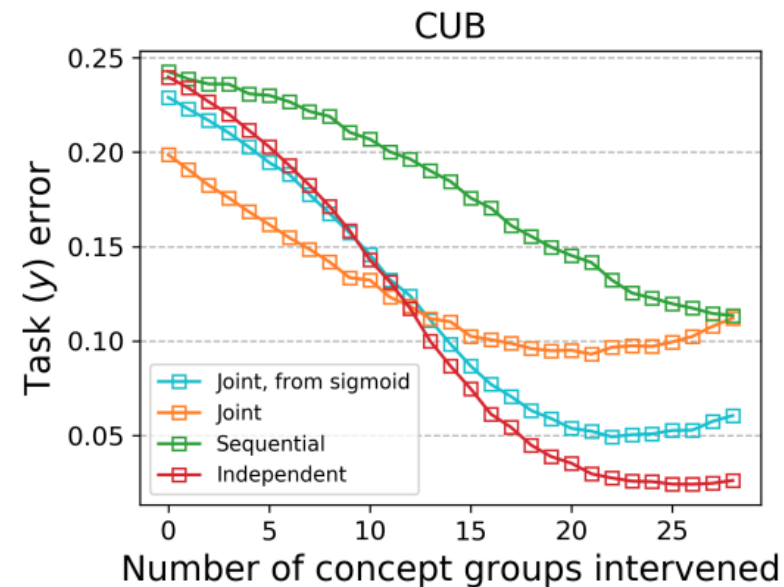
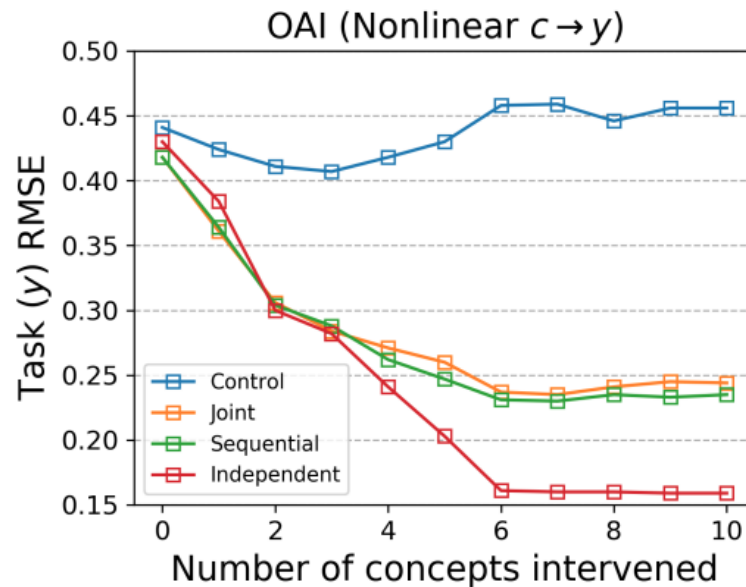
CONCEPT INTERVENTIONS



Our expert guest for today

CONCEPT INTERVENTIONS

As we intervene on more concepts, CBM's **test error goes down!**



ARE CBMS ALL WE NEED?

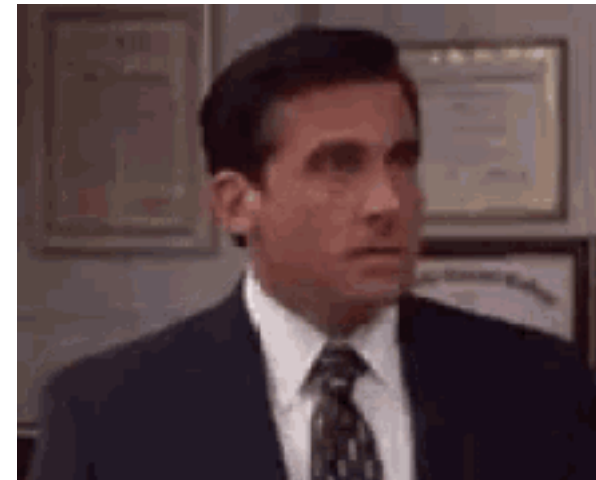
CBMs are great in a lot of ways:

1. They are simple to understand and provide **high-level explanations**.
2. They enable **test-time interventions** that improve their accuracy.
3. They are very **stable**, expressive and **easy to train**.

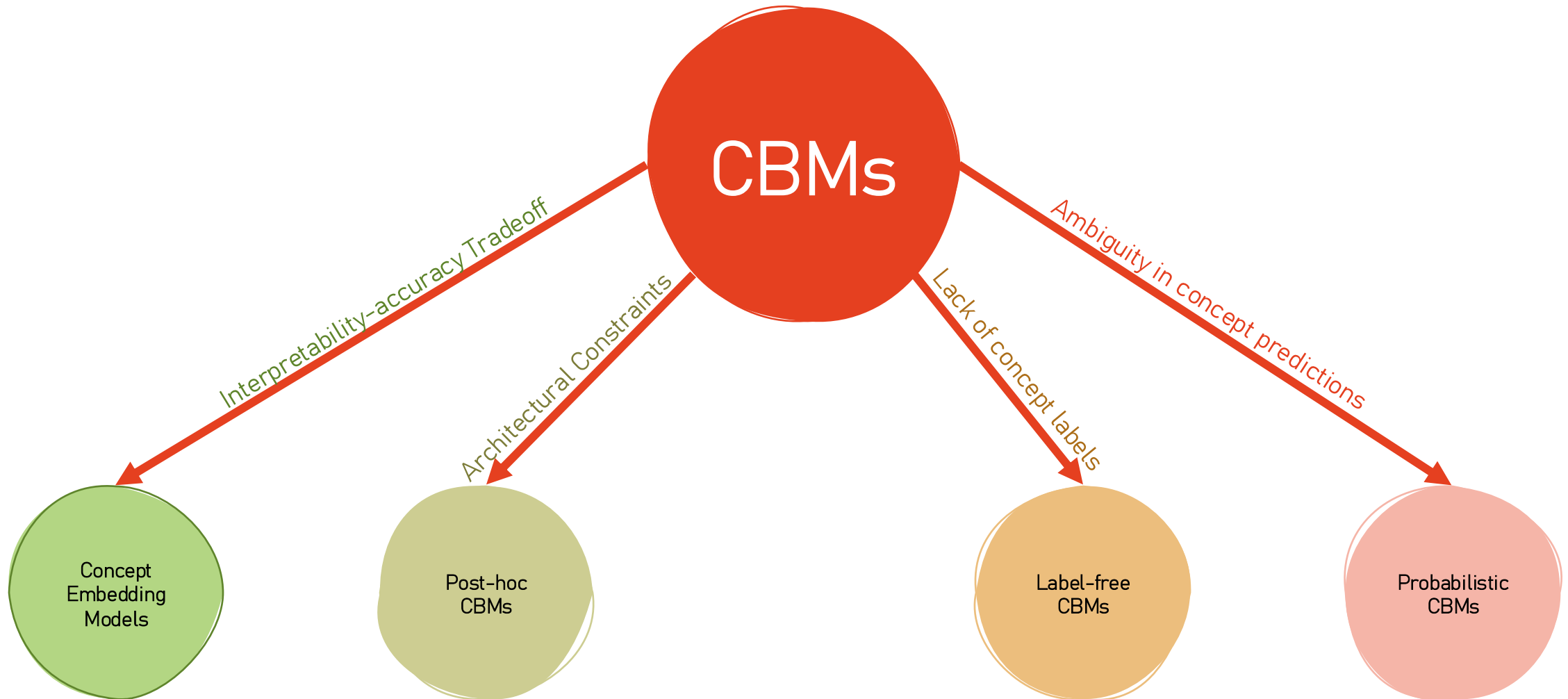
So, are we done?

Short Answer: No

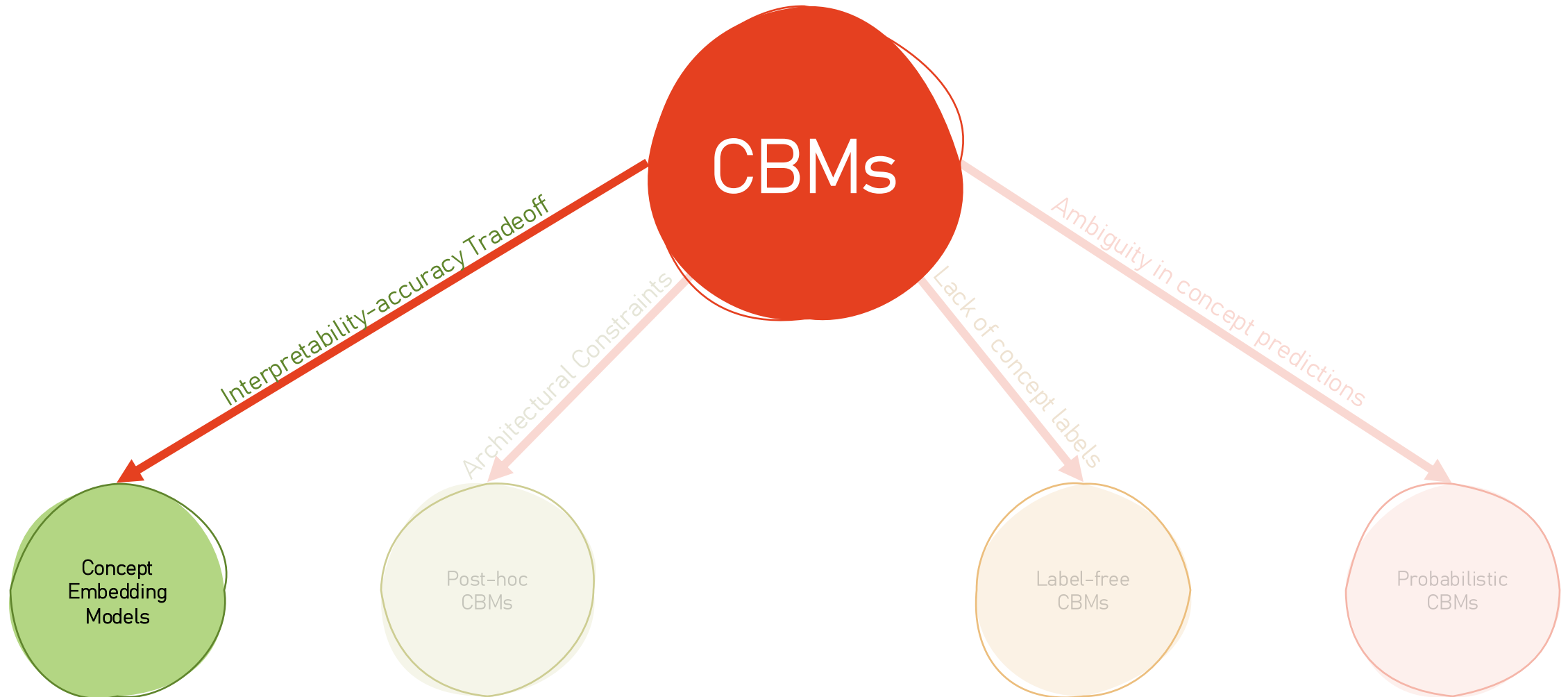
Long Answer:



INTRODUCING CBM'S FRIENDS



SPEED-DATING WITH CBM'S FRIENDS

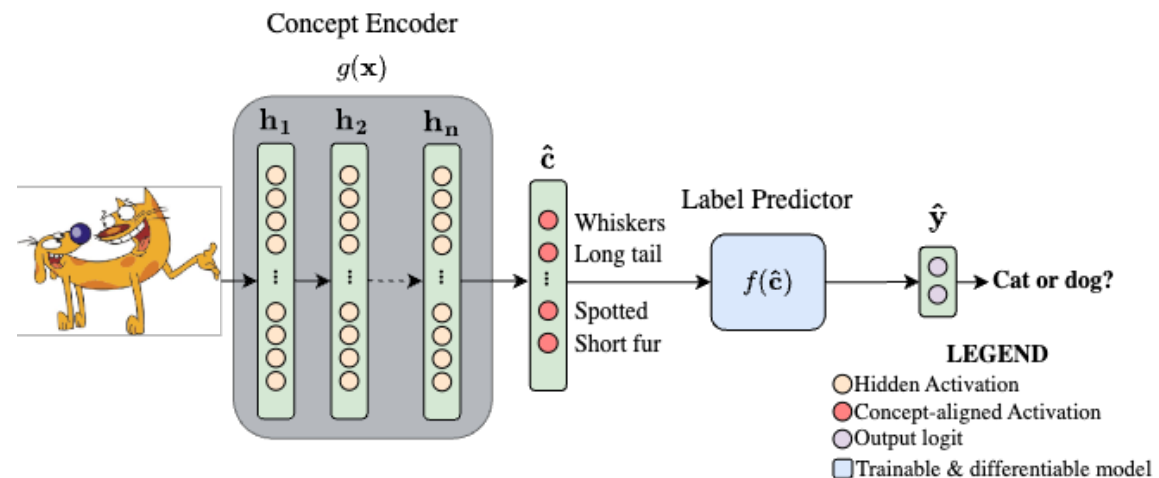
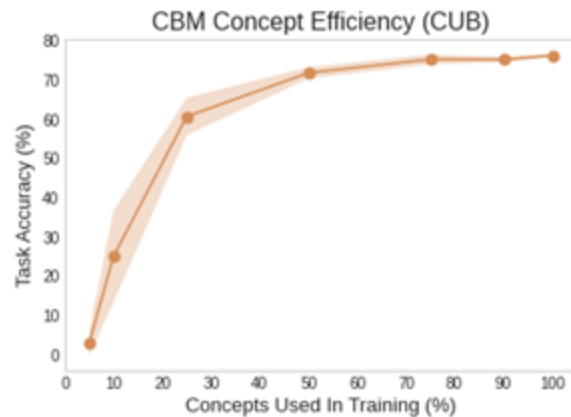


CONCEPT EMBEDDING MODELS

ESPINOSA ZARLENGA & BARBIERO ET AL. (NEURIPS 2022)

Limitation Being Addressed

Provided concepts need to be “complete” or else we observe a trade-off!



Q#1: which training regime do you think would perform better when concepts are incomplete?

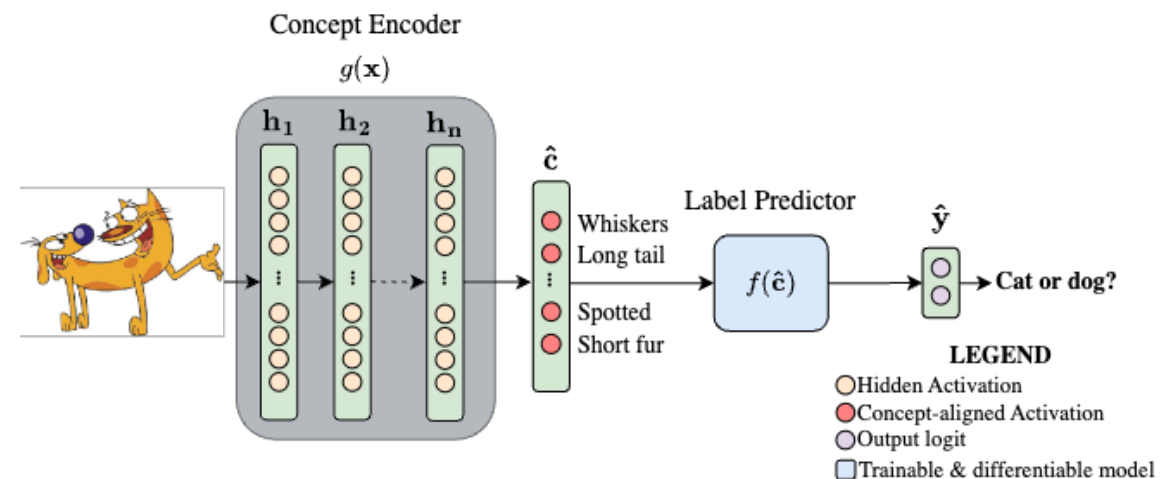
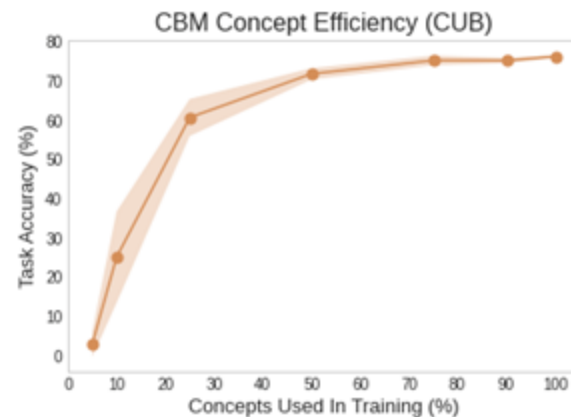


CONCEPT EMBEDDING MODELS

ESPINOSA ZARLENGA & BARBIERO ET AL. (NEURIPS 2022)

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Q#2: Why can't we just add a bypass from the input to the output?

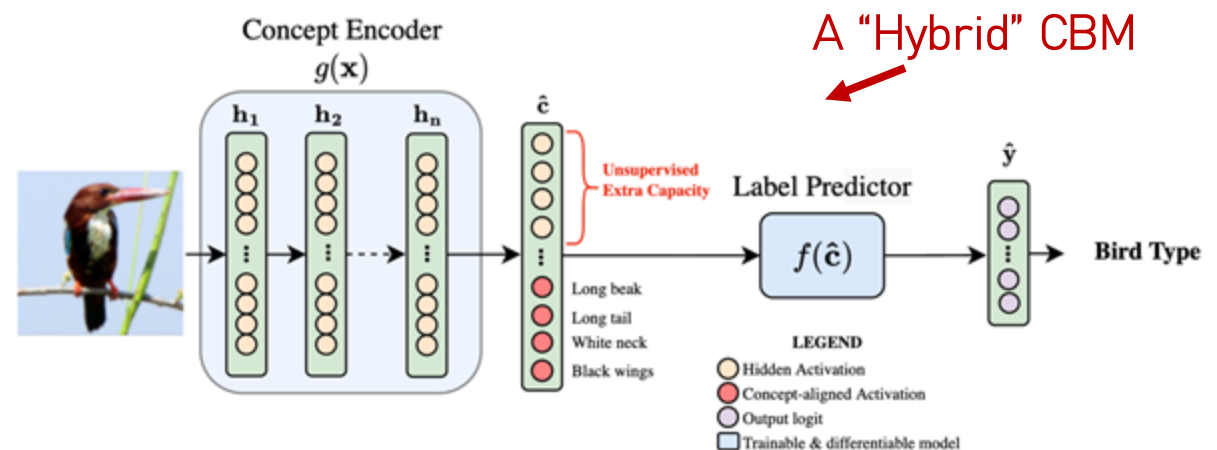
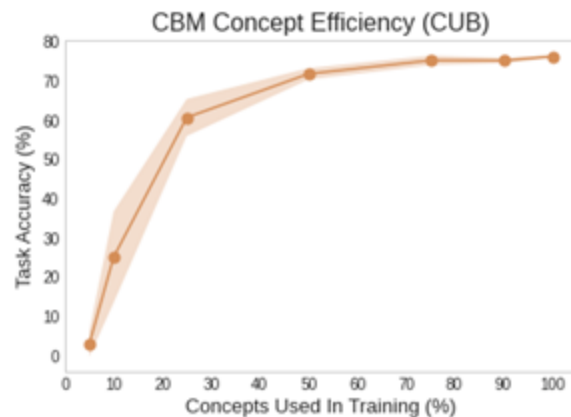


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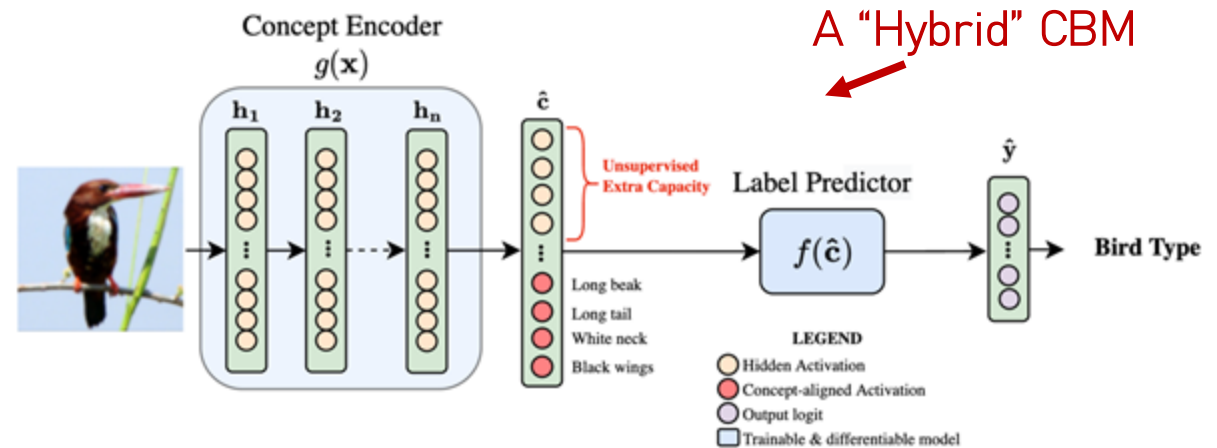
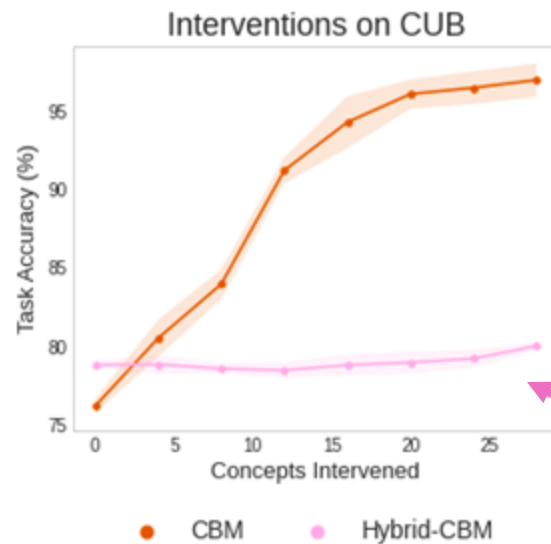


CONCEPT EMBEDDING MODELS

ESPINOSA ZARLENGA & BARBIERO ET AL. (NEURIPS 2022)

Limitation Being Addressed

Provided concepts need to be “complete” or else we observe a trade-off!



Interventions do not necessarily work with Hybrid CBMs!

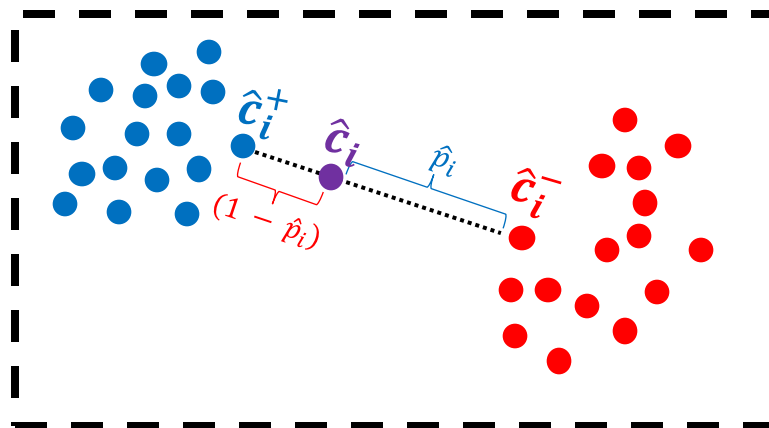


CONCEPT EMBEDDING MODELS

ESPINOSA ZARLENGA & BARBIERO ET AL. (NEURIPS 2022)

Proposed Solution

Learn **two high-dimensional embeddings for each concept** representing the concept when it is “**on**” and when it is “**off**”



Concept Embedding Space $\mathbb{R}^m$

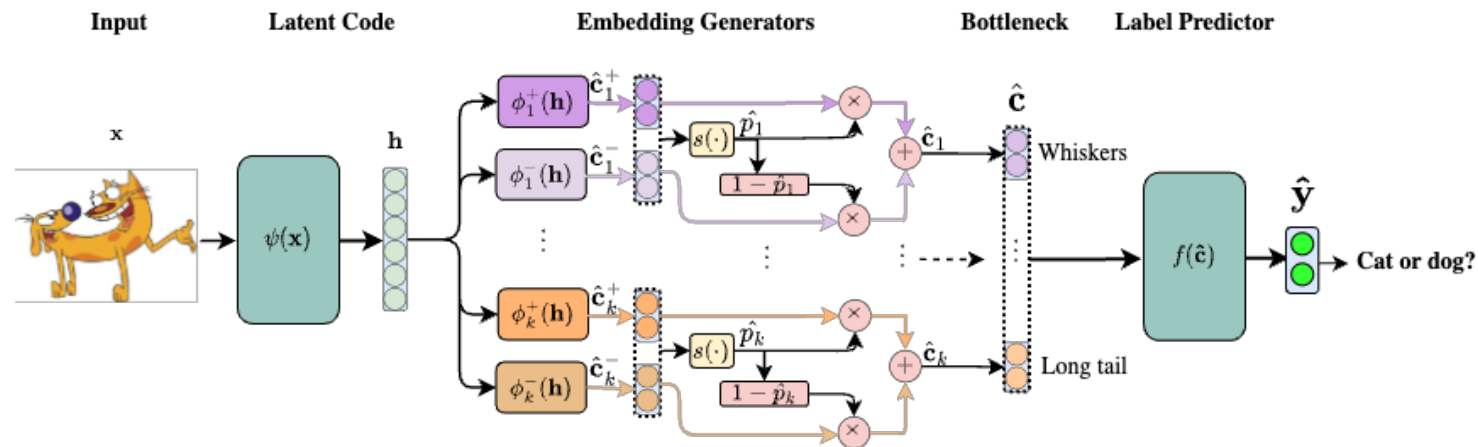


CONCEPT EMBEDDING MODELS

ESPINOSA ZARLENGA & BARBIERO ET AL. (NEURIPS 2022)

Proposed Solution

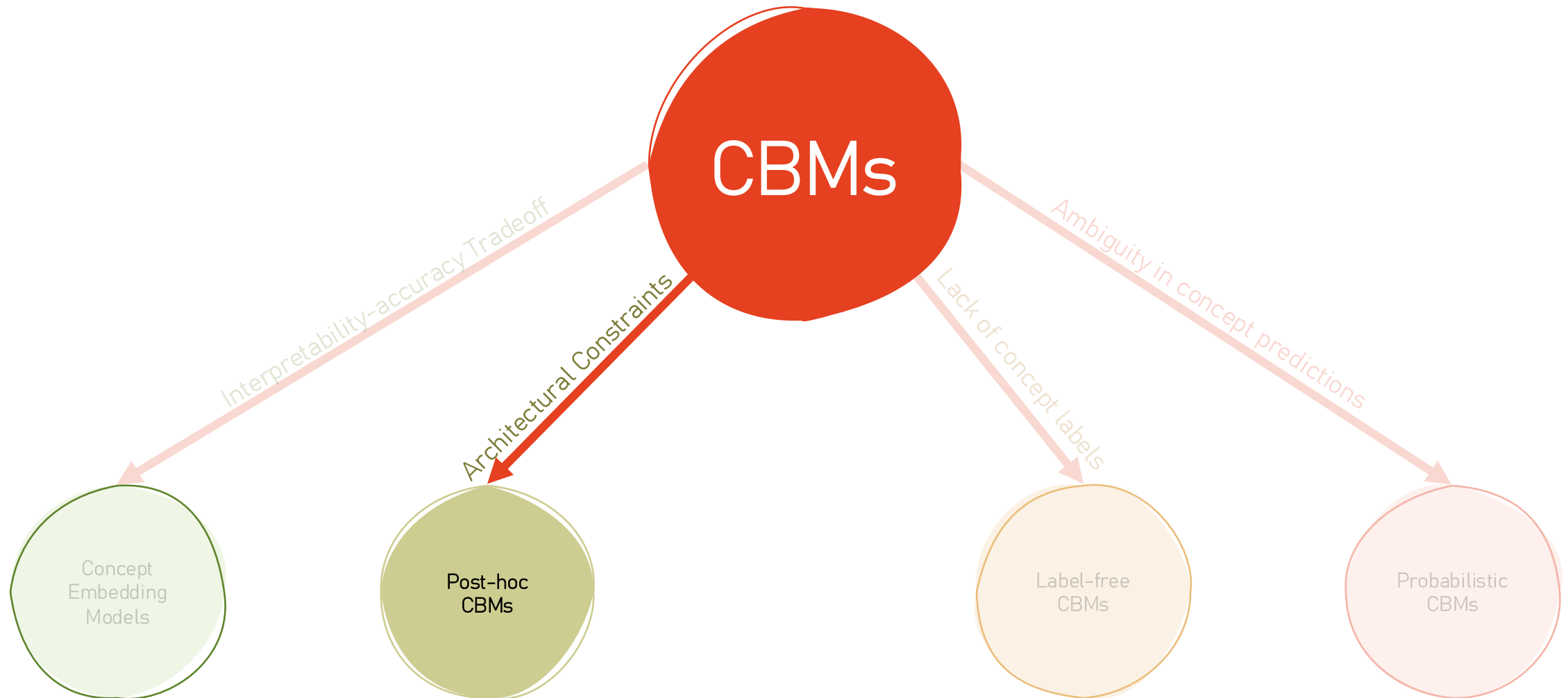
Learn **two high-dimensional embeddings for each concept** representing the concept when it is “**on**” and when it is “**off**”



Mix the two embeddings based on their **predicted probability**: $\hat{c}_i = \hat{p}_i \mathbf{c}_i^{(+)} + (1 - \hat{p}_i) \mathbf{c}_i^{(-)}$



SPEED-DATING WITH CBM'S FRIENDS

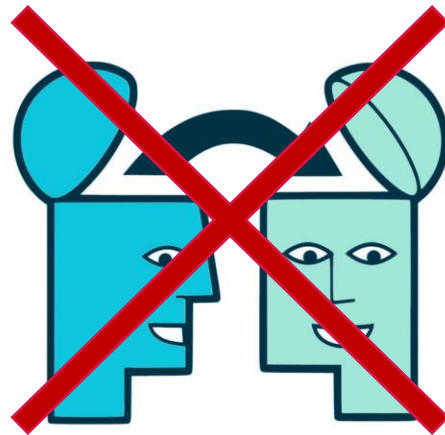


POST-HOC CBMS

YUKSEKGONUL ET AL. (ICLR 2023)

Limitation Being Addressed

Training a CBMs requires training from scratch, leading to **significant constraints and architectural changes**, and it **requires all training samples to be concept annotated!**



POST-HOC CBMS

YUKSEKGONUL ET AL. (ICLR 2023)

Proposed Solution

Can we align a layer in a pre-trained model to concept scores obtained using T-CAV?

This would allow us to just finetune a single part of a pre-trained model using concept annotations in potentially distinct datasets!

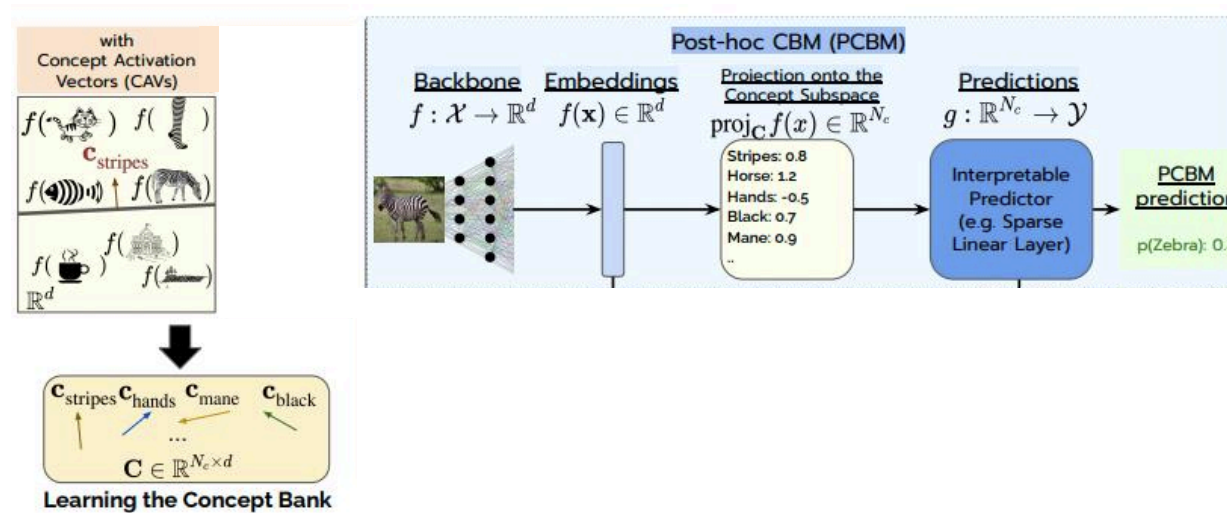


POST-HOC CBMS

YUKSEKGONUL ET AL. (ICLR 2023)

Proposed Solution

Align a layer in a **pre-trained model** to concept scores obtained using **T-CAV**



Make the final prediction with an **interpretable predictor**

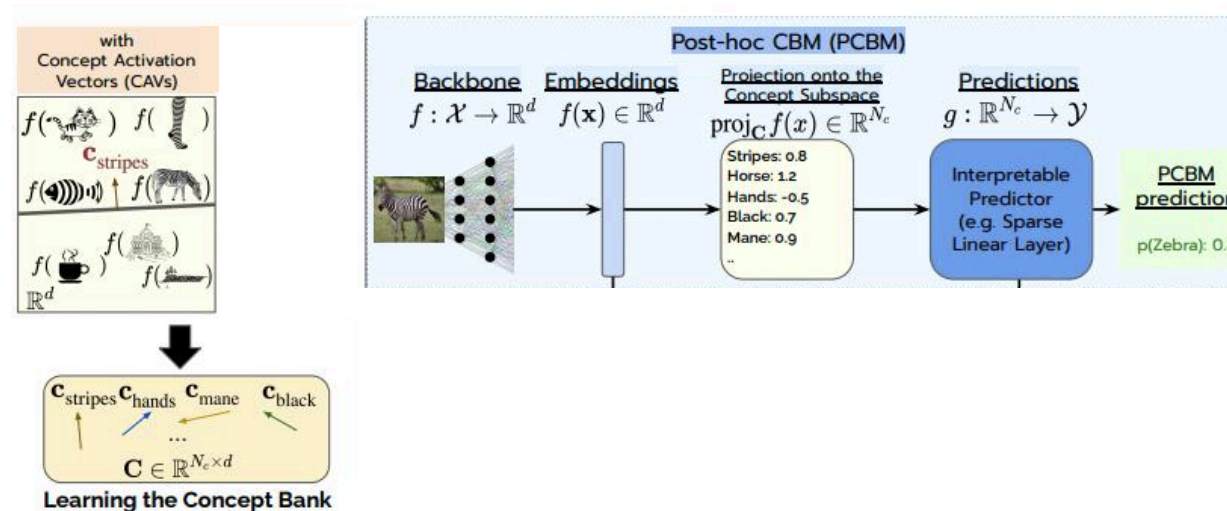


POST-HOC CBMS

YUKSEKGONUL ET AL. (ICLR 2023)

Proposed Solution

Align a layer in a **pre-trained model** to concept scores obtained using **T-CAV**



Make the final prediction with an **interpretable predictor**

Question: do you see any issues with this architecture? (hint: think of our previous discussion)

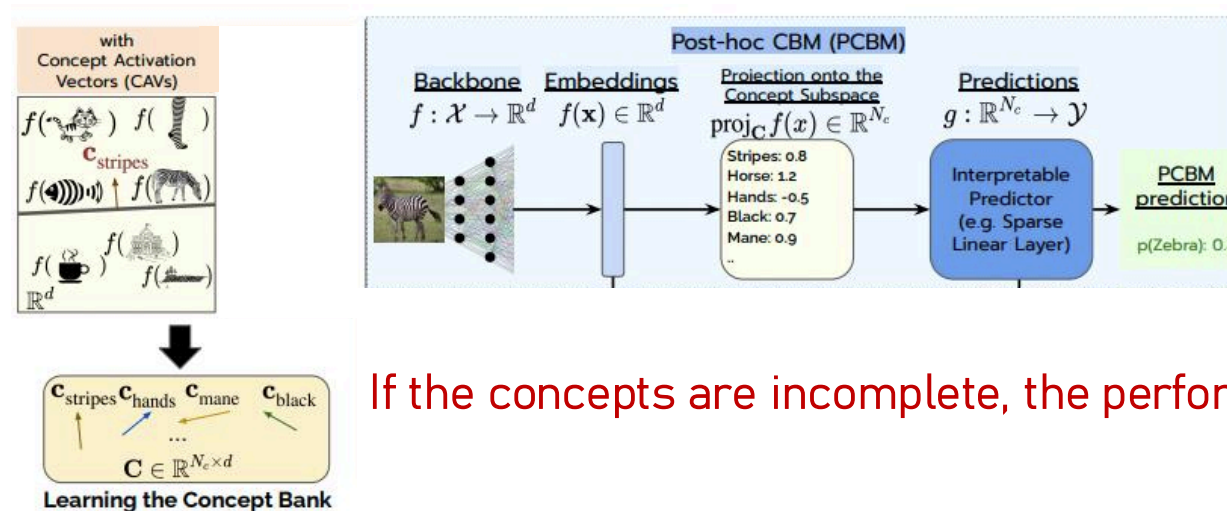


POST-HOC CBMS

YUKSEKGONUL ET AL. (ICLR 2023)

Proposed Solution

Align a layer in a **pre-trained model** to concept scores obtained using **T-CAV**



Make the final prediction with an **interpretable predictor**

If the concepts are incomplete, the performance will drop significantly!

Question: do you see any issues with this architecture? (hint: think of our previous discussion)

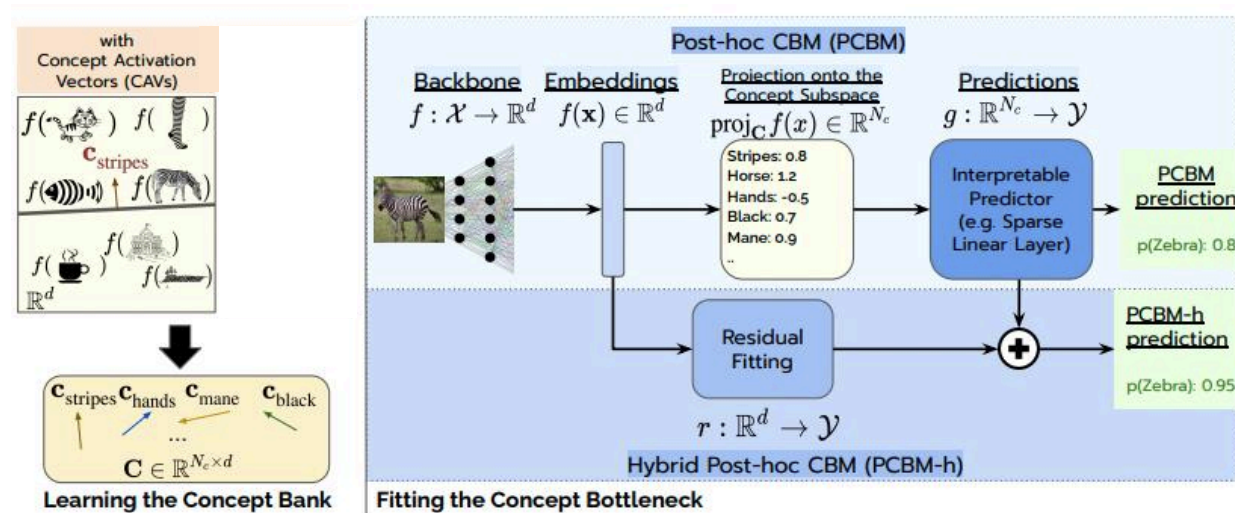


POST-HOC CBMS

YUKSEKGONUL ET AL. (ICLR 2023)

Proposed Solution

Align a layer in a pre-trained model to concept scores obtained using T-CAV



A residual model can be fitted if the concept bank is incomplete!

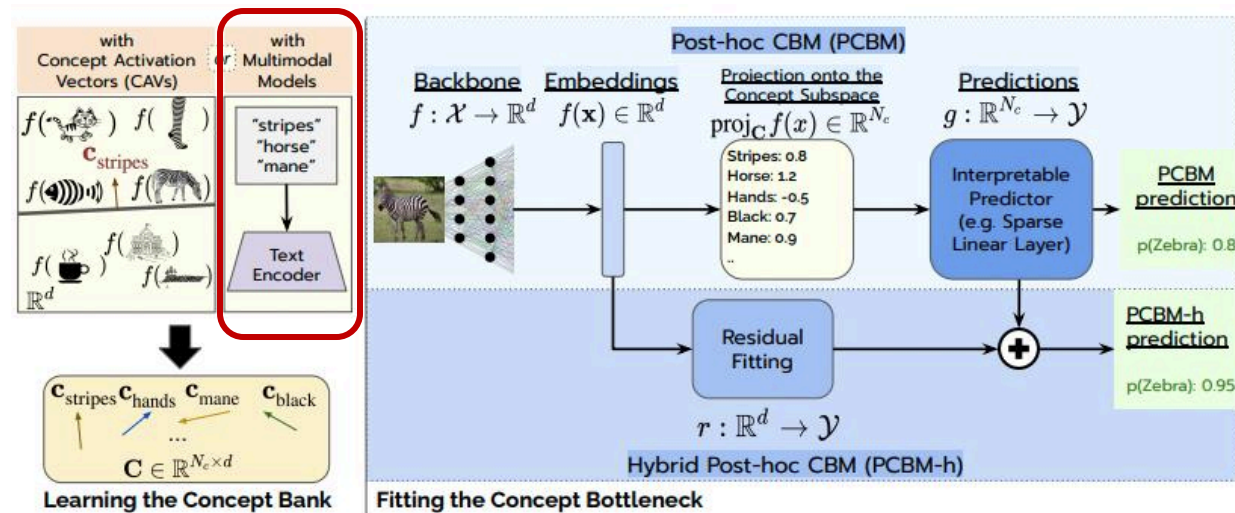


POST-HOC CBMS

YUKSEKGONUL ET AL. (ICLR 2023)

Proposed Solution

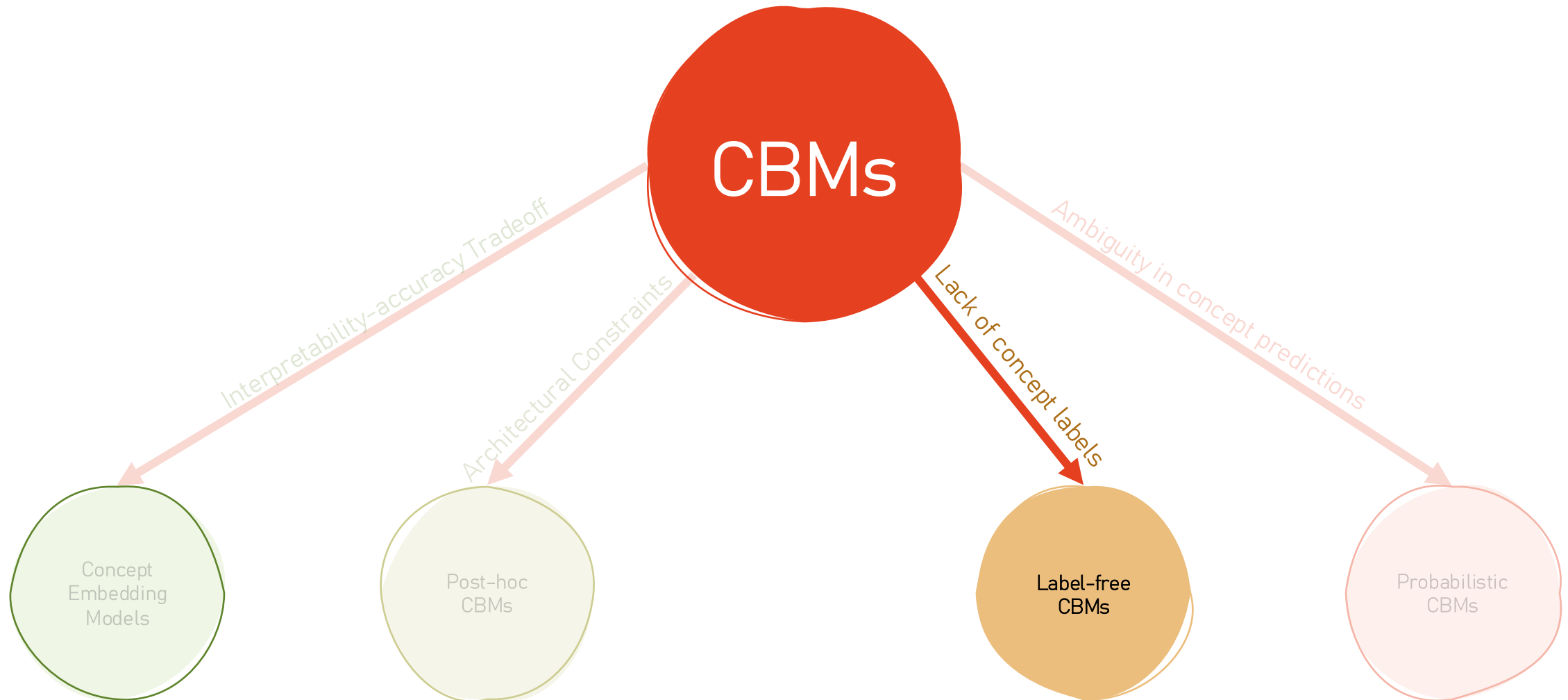
Align a layer in a pre-trained model to concept scores obtained using T-CAV



CAVs can be learnt using language-based concepts together with a multimodal model to learn CAVs!



SPEED-DATING WITH CBM'S FRIENDS

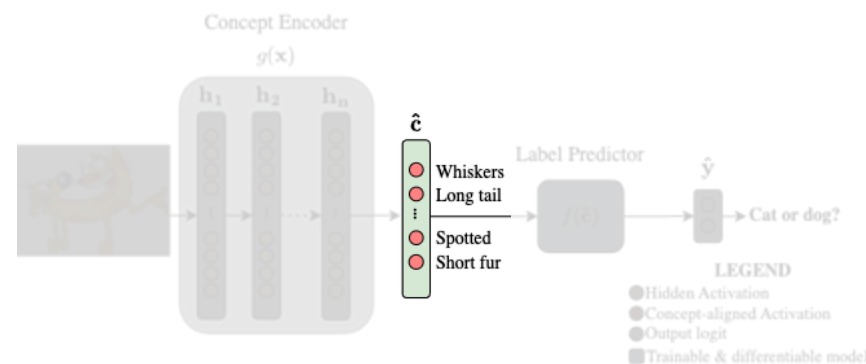


LABEL-FREE CBMS

OIKARINEN ET AL. (ICLR 2023)

Limitation Being Addressed

CBMs and CEMs **require some known concepts** or we have no bottleneck at all!



And post-hoc CBMs still **require one to know which concepts are potentially useful for a downstream task!**



LABEL-FREE CBMS

OIKARINEN ET AL. (ICLR 2023)

Proposed Solution

Why not **simply ask GPT** for a set of useful concepts for a specific class?



"List the most important features for recognizing something as a {class}:"



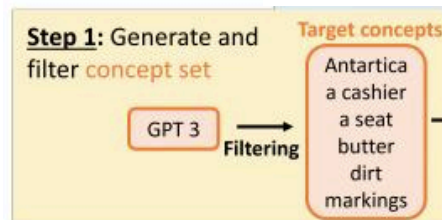
LABEL-FREE CBMS

OIKARINEN ET AL. (ICLR 2023)

Proposed Solution

Step 1: generate a concept set by "asking" an LLM

Label-free CBM

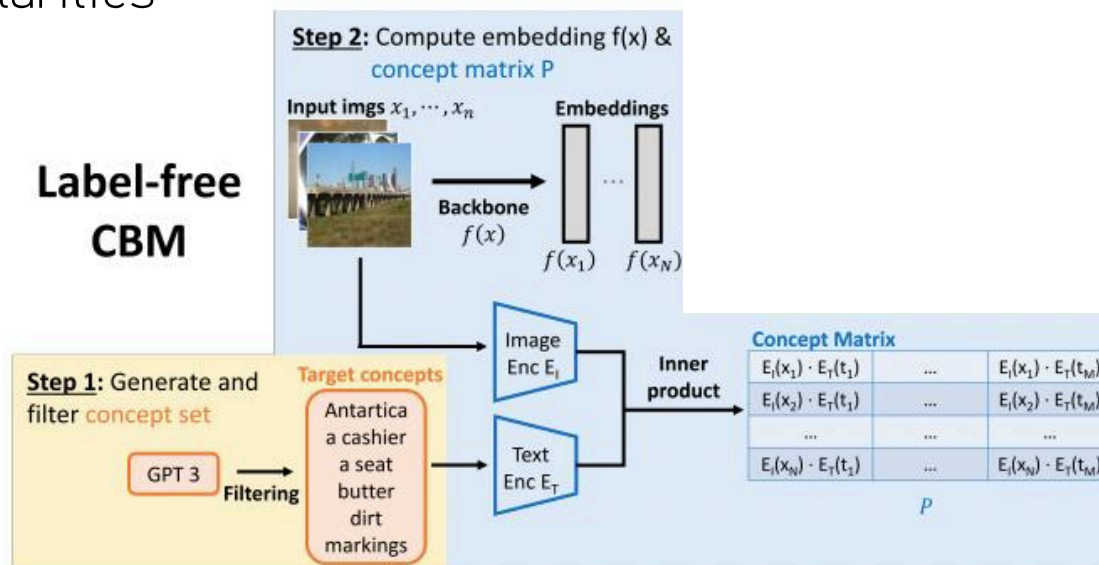


LABEL-FREE CBMS

OIKARINEN ET AL. (ICLR 2023)

Proposed Solution

Step 2: Map samples to an embedding space using a VLM (e.g., CLIP) and compute concept similarities

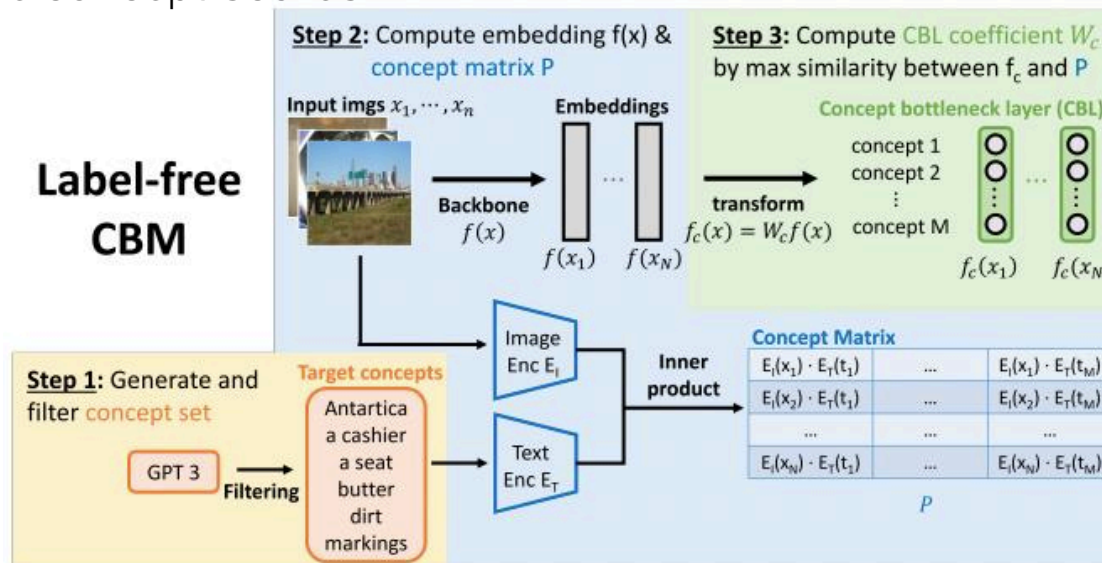


LABEL-FREE CBMS

OIKARINEN ET AL. (ICLR 2023)

Proposed Solution

Step 3: Learn a **linear mapping** between a **backbone's embeddings** and a **vector whose activations are aligned** with the concept scores



W_c is trained to **maximise the correlation** between its i -th output and the i -th concept's scores

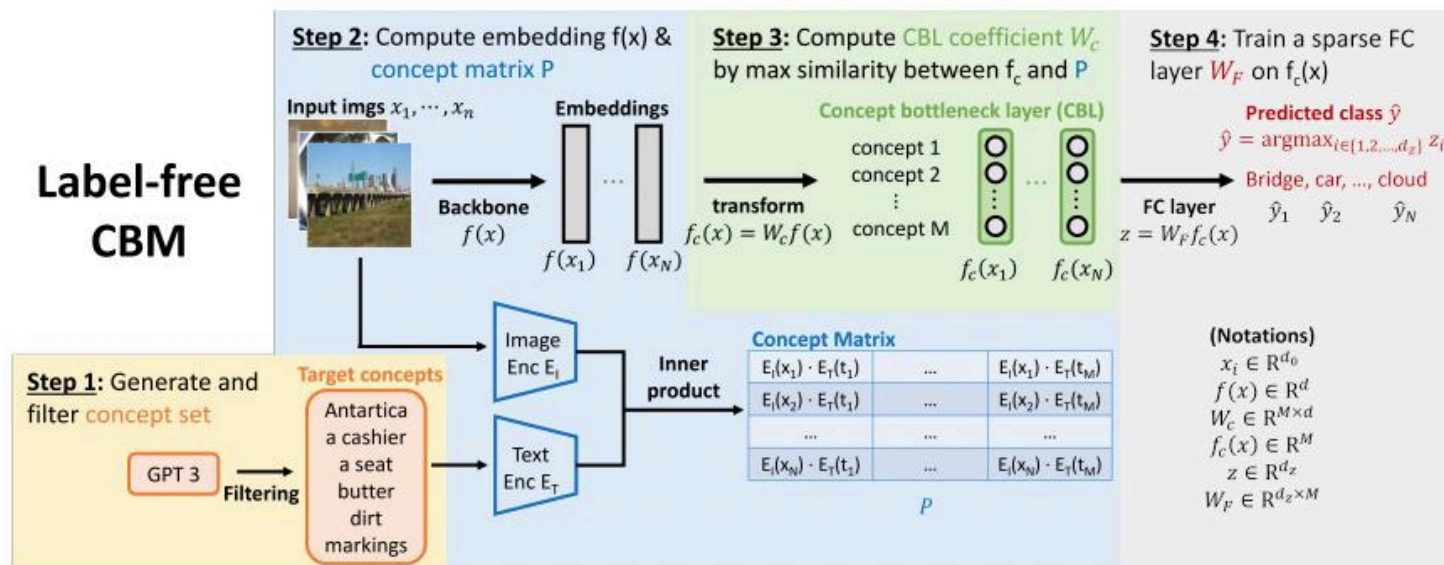


LABEL-FREE CBMS

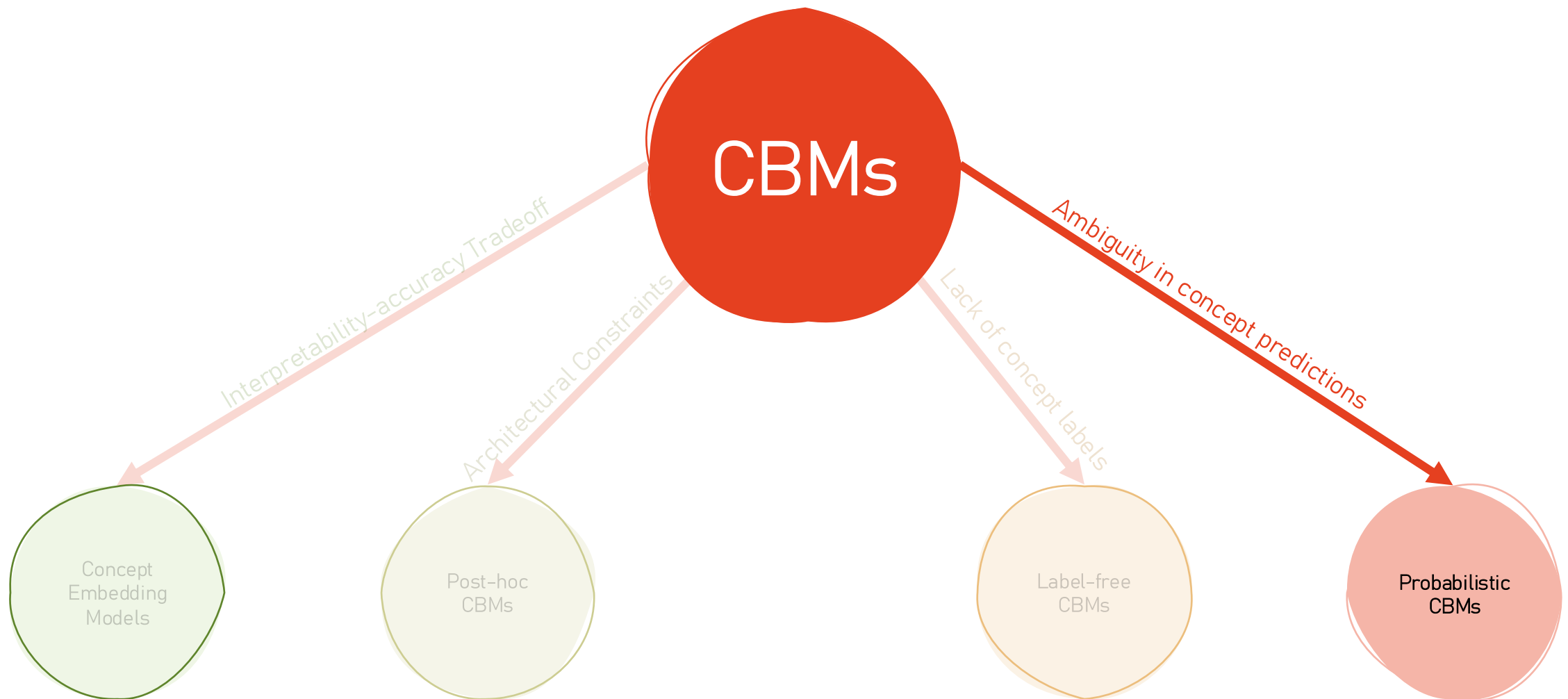
OIKARINEN ET AL. (ICLR 2023)

Proposed Solution

Step 4: Train a sparse (interpretable) model to map predicted concept scores to tasks



SPEED-DATING WITH CBM'S FRIENDS



PROBABILISTIC CBMS

KIM ET AL. (ICML 2023)

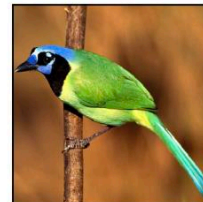
Limitation Being Addressed

CBMs **must predict concepts** for all samples even they are **ambiguous**

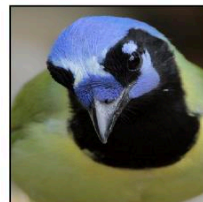
Class: Green Jay

Concepts:

forehead color: blue
throat color: black
belly color: yellow
tail pattern: solid



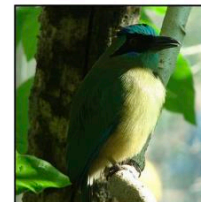
Diverse visual contexts



ambiguity in tail



ambiguity in belly



ambiguity in color

The cross-entropy loss **does not encourage the concept predictor to be uncertain**



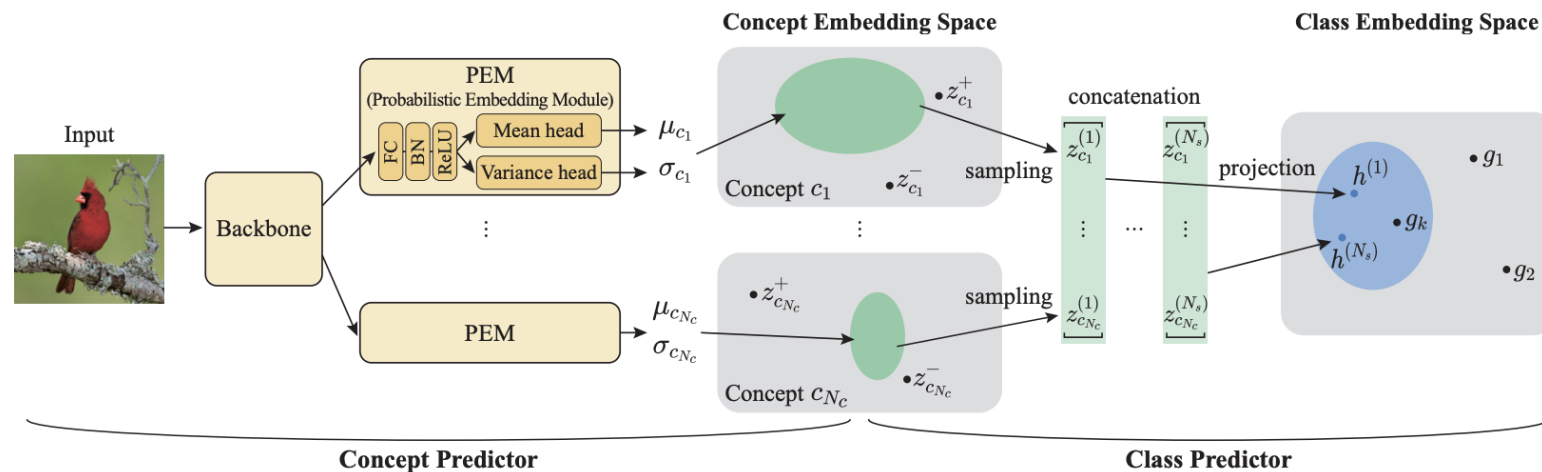
PROBABILISTIC CBMS

KIM ET AL. (ICML 2023)

Proposed Solution

Use **probabilistic embeddings** that enable **uncertainty estimation** of each concept!

Learn a **distribution over concept embeddings** and use its **variance** to estimate **uncertainty**

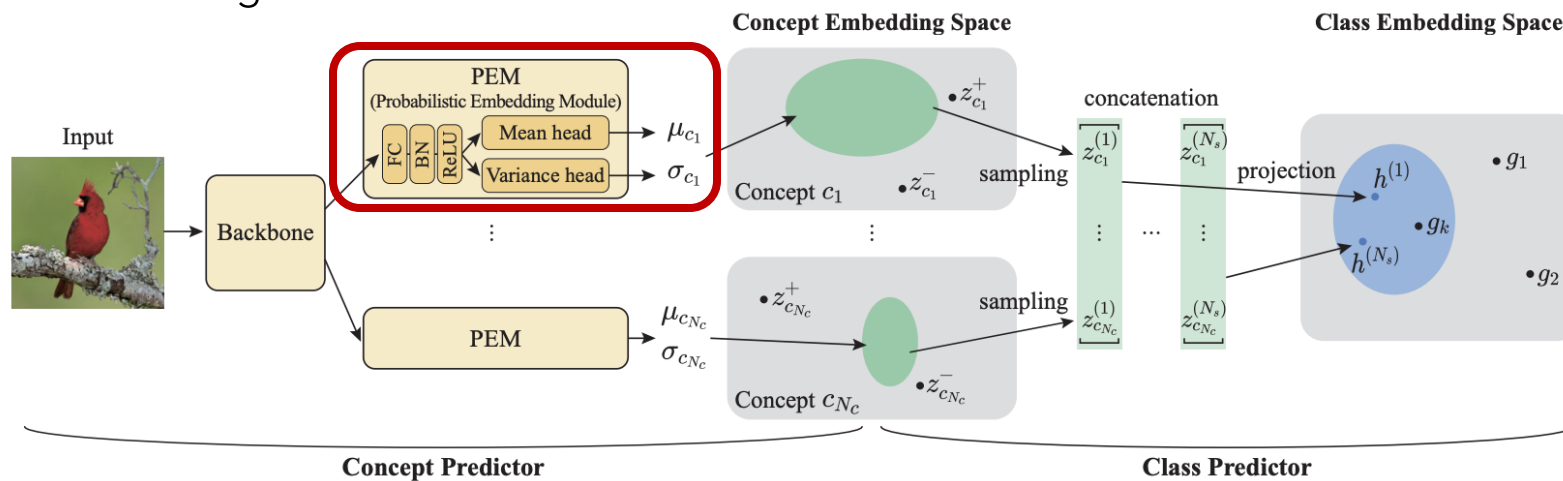


PROBABILISTIC CBMS

KIM ET AL. (ICML 2023)

Proposed Solution

Each **Probabilistic Embedding Module (PEM)** generates a **mean** μ_{c_i} and a **variance** σ_{c_i} for the concept embedding



$$p(z_c|x) \sim \mathcal{N}(\mu_c, \text{diag}(\sigma_c))$$

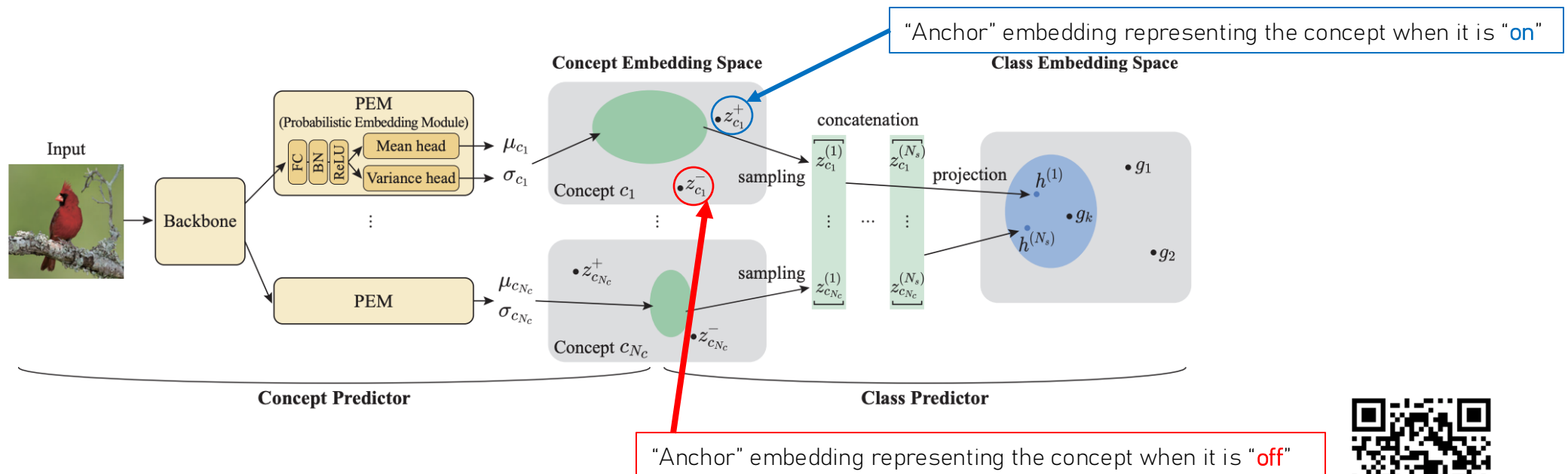


PROBABILISTIC CBMS

KIM ET AL. (ICML 2023)

Proposed Solution

We learn a set of fixed **anchor embeddings** representing the concept when it is **on** vs **off**

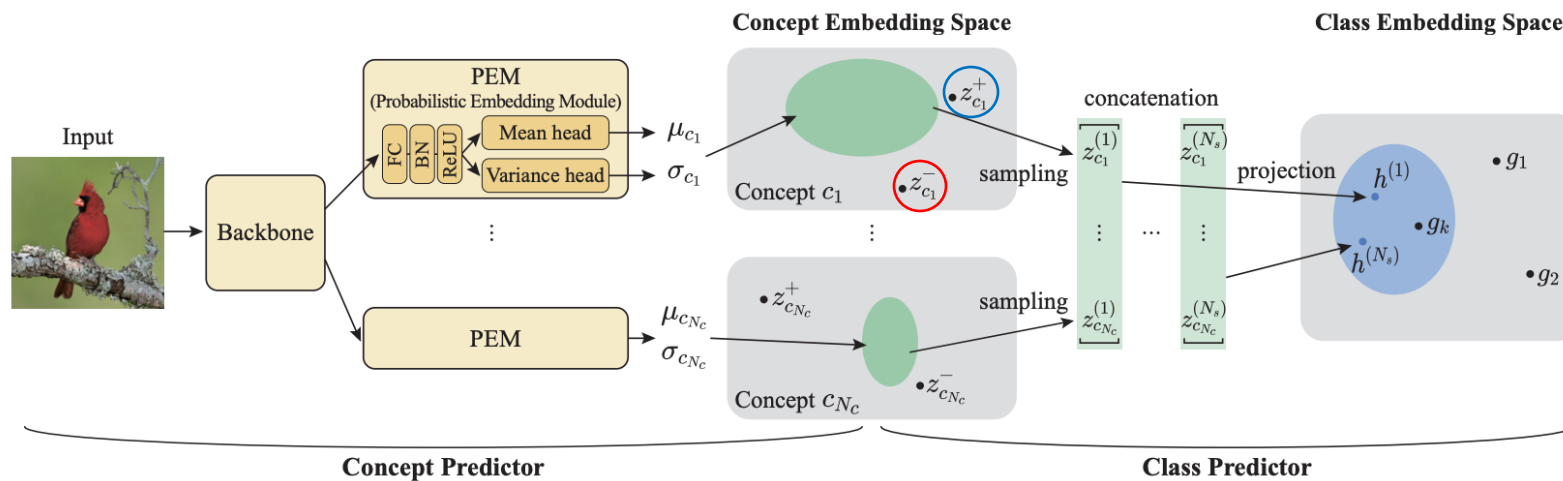


PROBABILISTIC CBMS

KIM ET AL. (ICML 2023)

Proposed Solution

The **distance** from the sampled embedding to each anchor can be used to **predict a concept**!



$$p(c = 1 | z_c) = \sigma \left(a \left(\|z_c - z_c^-\|_2 - \|z_c - z_c^+\|_2 \right) \right)$$

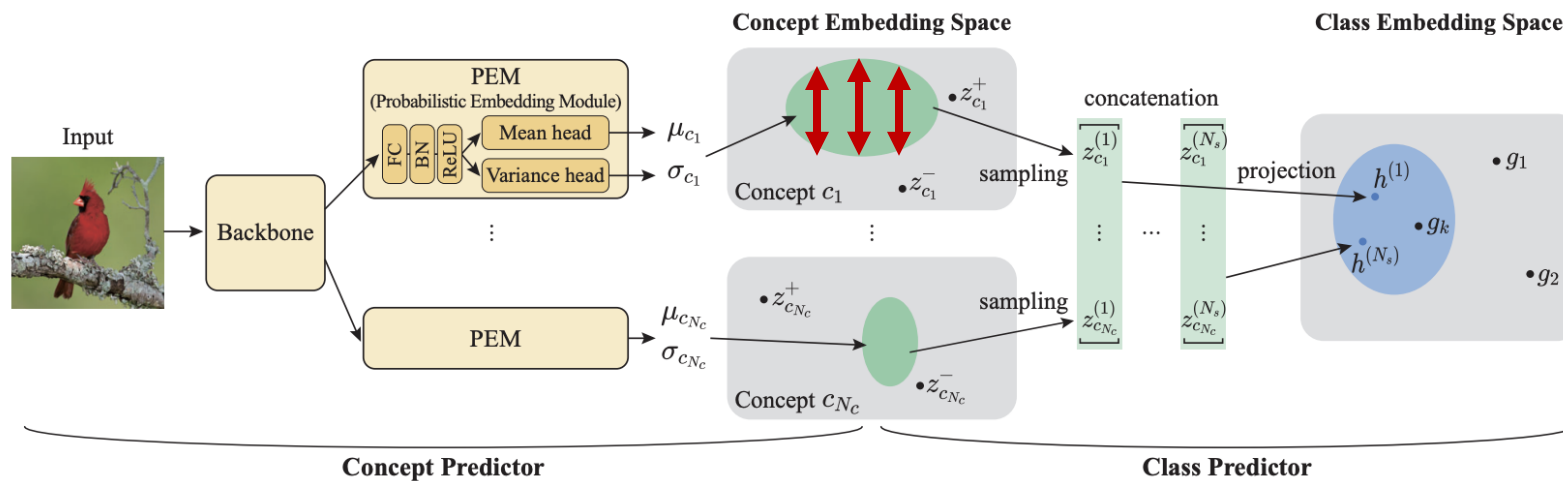


PROBABILISTIC CBMS

KIM ET AL. (ICML 2023)

Proposed Solution

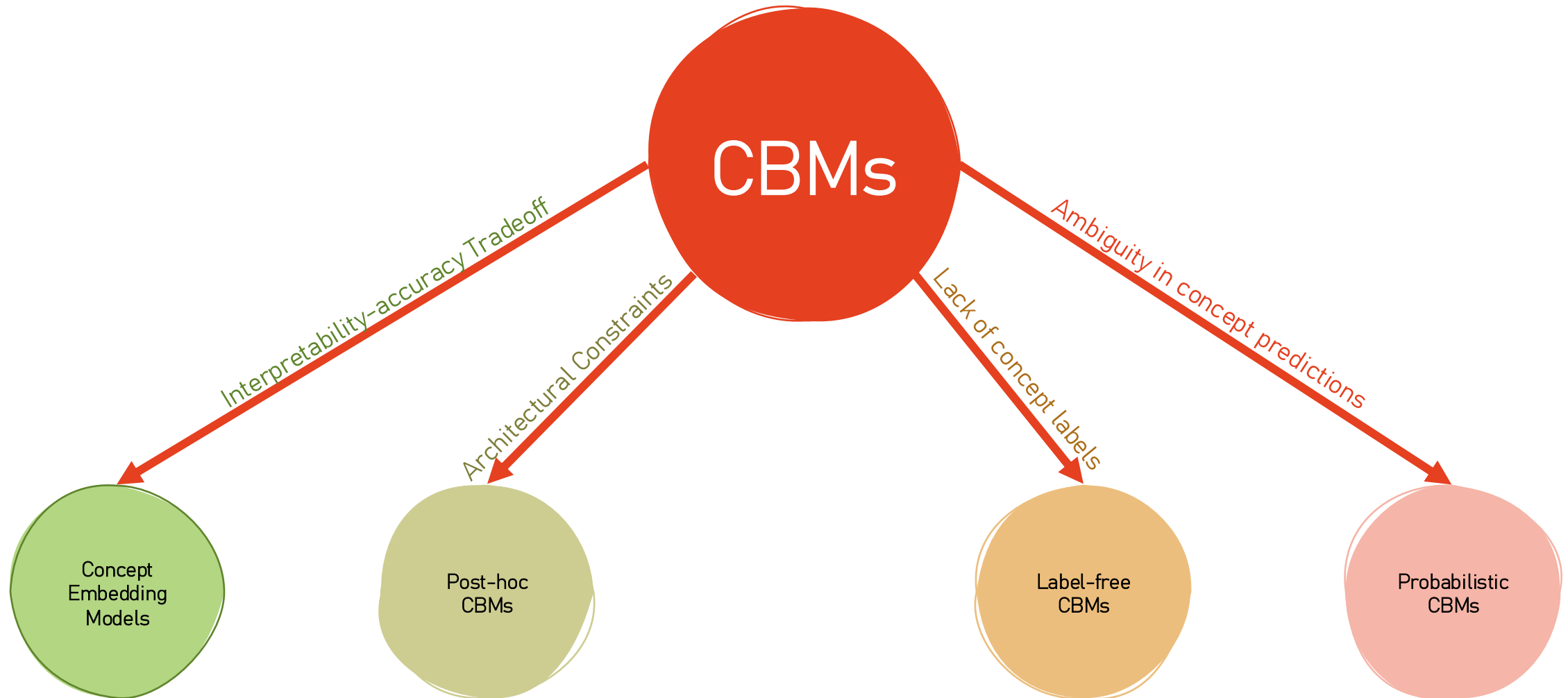
A **concept's distribution's volume** can be used to **quantify its uncertainty**!



As **embeddings are modelled as Gaussians**, this is the **determinant of the covariance**!



END OF OUR SPEED DATING!



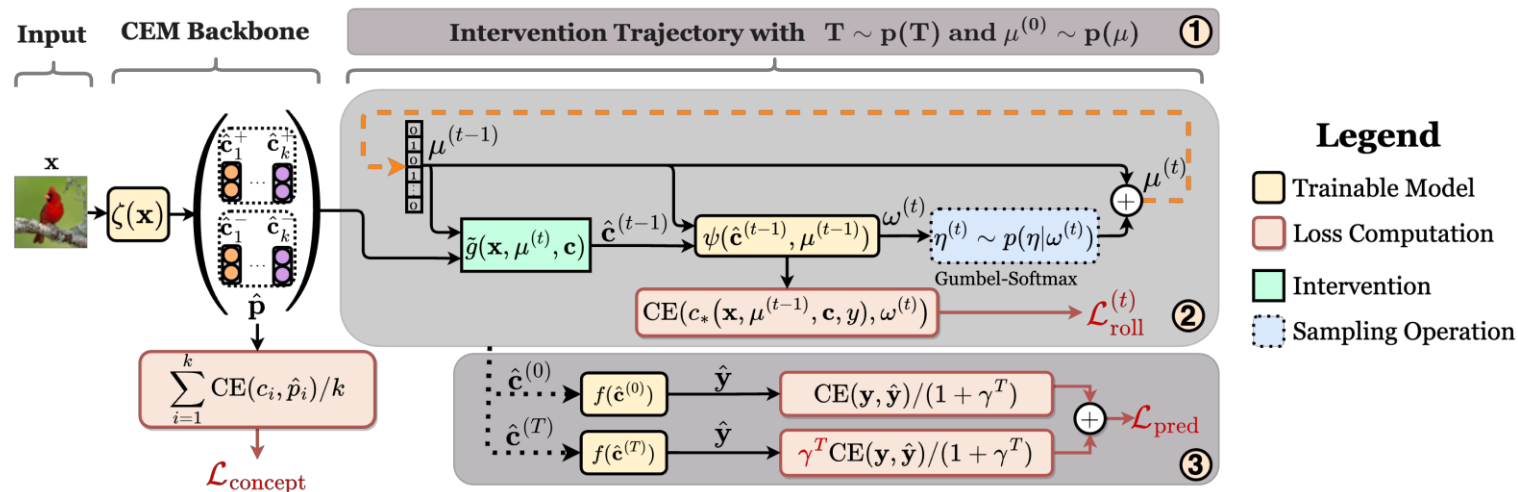
NEW DIRECTIONS

CBMs have become **very popular in XAI** with several active **areas of research**:

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1. Improving the effect of **concept interventions** (a topic of personal importance)



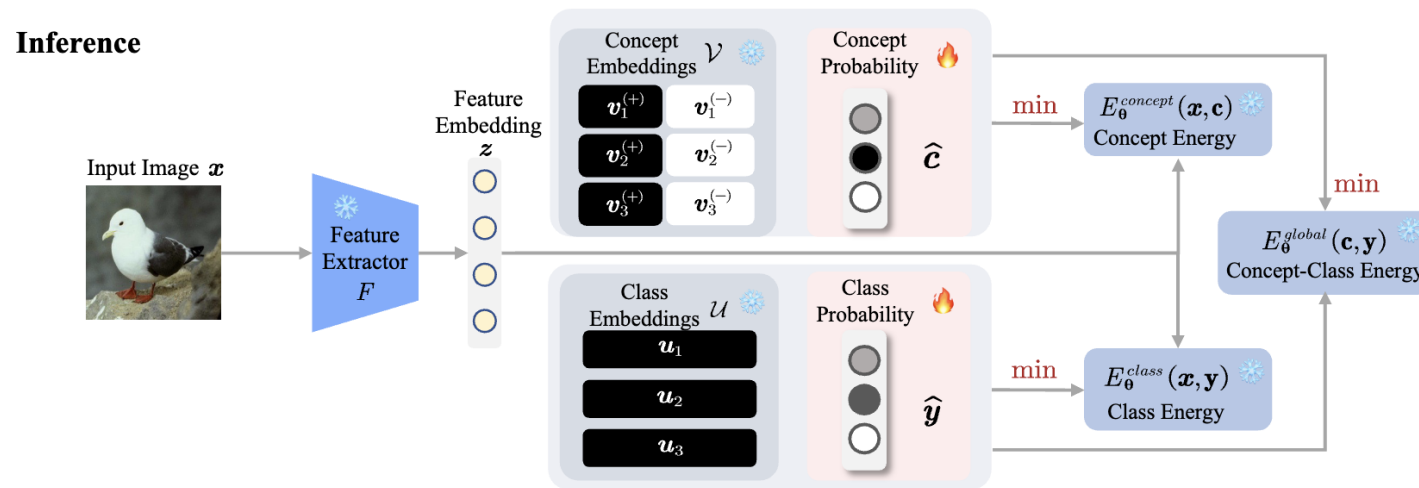
Intervention-aware Concept Embedding Models (Espinosa Zarlenga et al., 2023)



NEW DIRECTIONS

CBMs have become **very popular in XAI** with several active **areas of research**:

1. Generalising **Concept Interventions** (a topic of personal importance)
2. Understanding how to better **model concept-to-concept relationships**



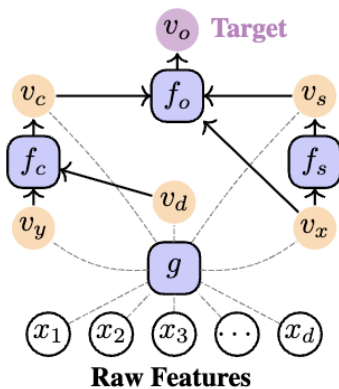
Energy-based Concept Bottleneck Models (Xu et al., 2024)



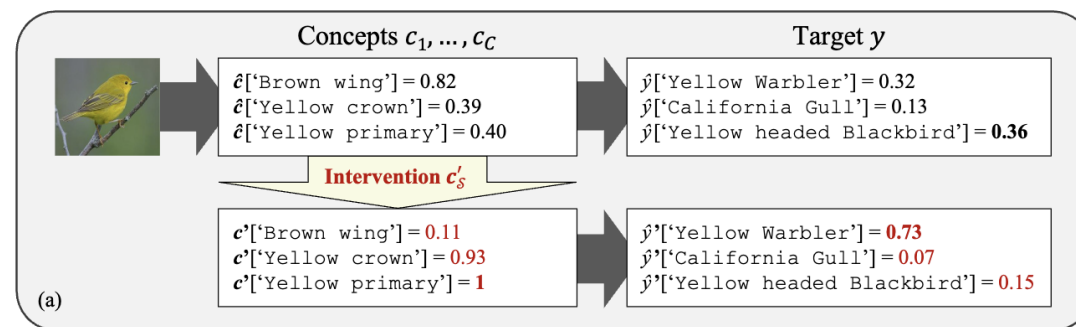
NEW DIRECTIONS

CBMs have become **very popular in XAI** with several active **areas of research**:

1. Generalising **Concept Interventions** (a topic of personal importance)
2. Understanding how to better **model concept-to-concept relationships**
3. Exploring the relationship between concepts, tasks, and **causality**



Causal Concept Graph Models (Dominici et al.)



Stochastic CBMs (Vandenhirtz et al.)

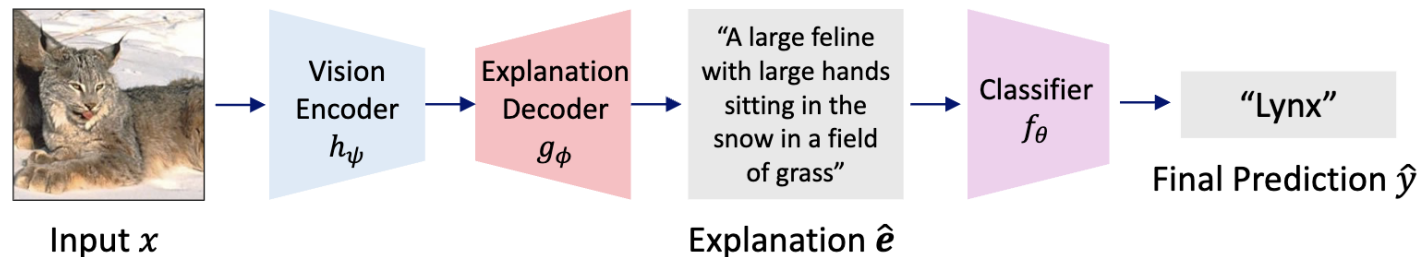
[1] Dominici and Barbiero et al. "Causal Concept Embedding Models: Beyond Causal Opacity in Deep Learning." ICLR (2025)

[2] Vandenhirtz and Laguna et al. "Stochastic concept bottleneck models." NeurIPS (2024)..

NEW DIRECTIONS

CBMs have become **very popular in XAI** with several active **areas of research**:

1. Generalising **Concept Interventions** (a topic of personal importance)
2. Understanding how to better **model concept-to-concept relationships**
3. Exploring the relationship between concepts, tasks, and **causality**
4. Producing entirely **language-based bottlenecks** (very recent!)



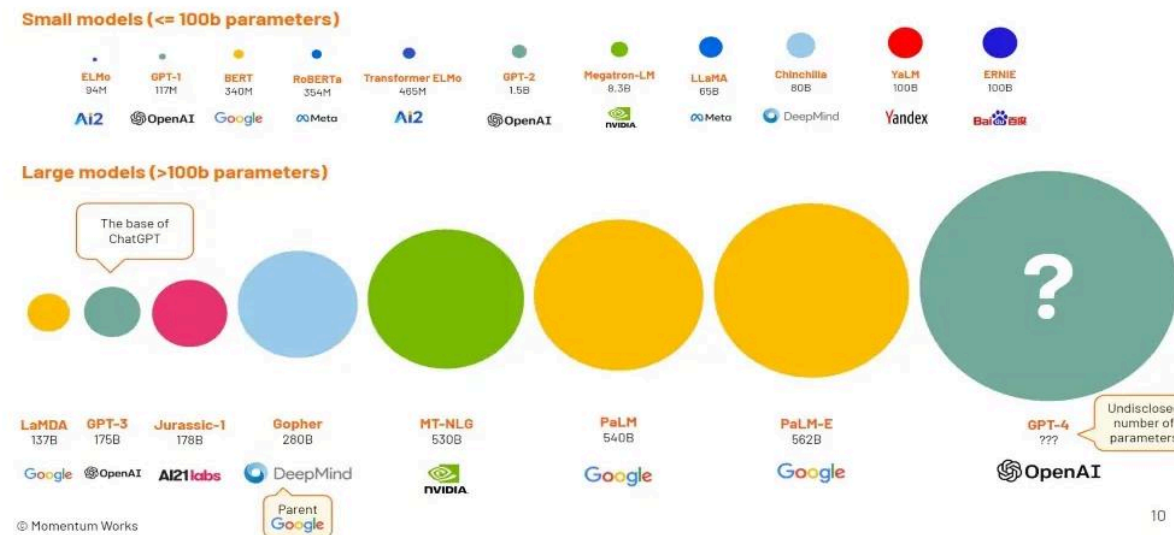
Explanation Bottleneck Models (Yamaguchi et al.)



ARE CBMS THE SOLUTION?

Maybe... but they still have some clear open questions:

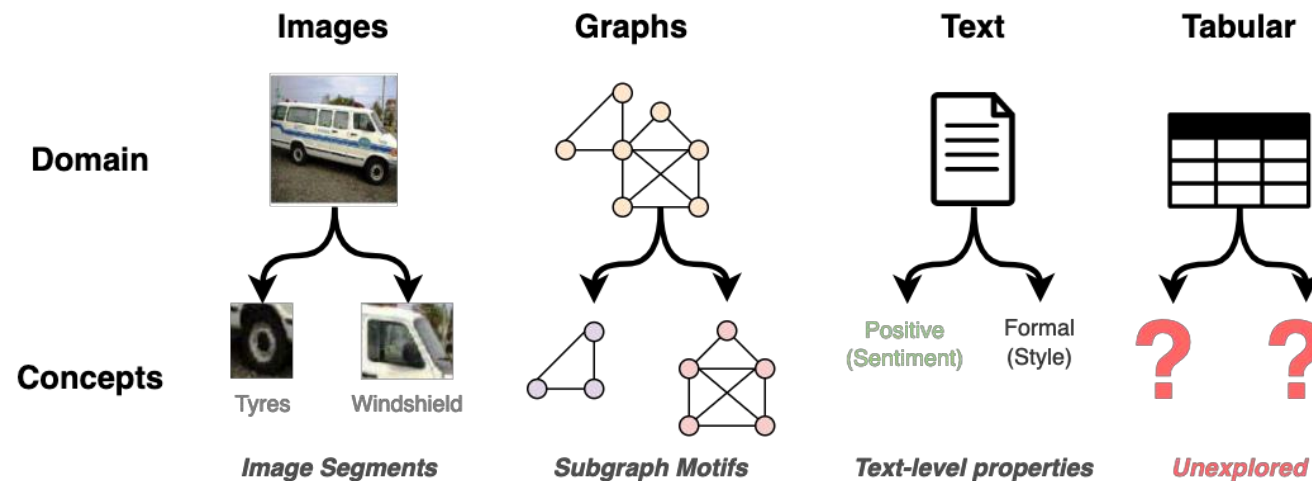
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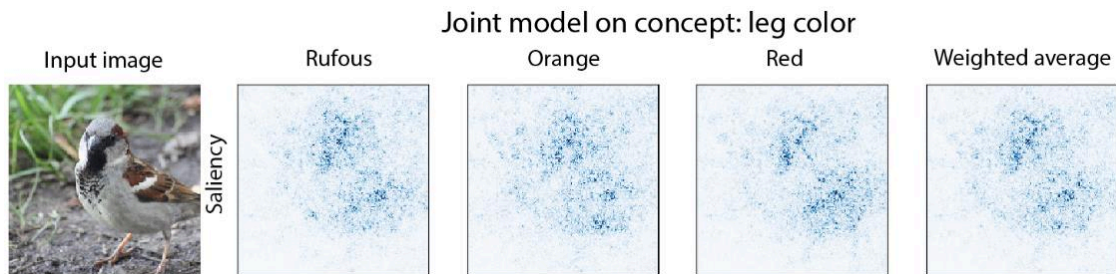
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2. Concepts are **ill-defined in a lot of domains** (e.g., tabular data, speech, etc)
3. Concepts are still predicted with black-box models, leading to accidental **leakage**



ARE CBMS THE SOLUTION?

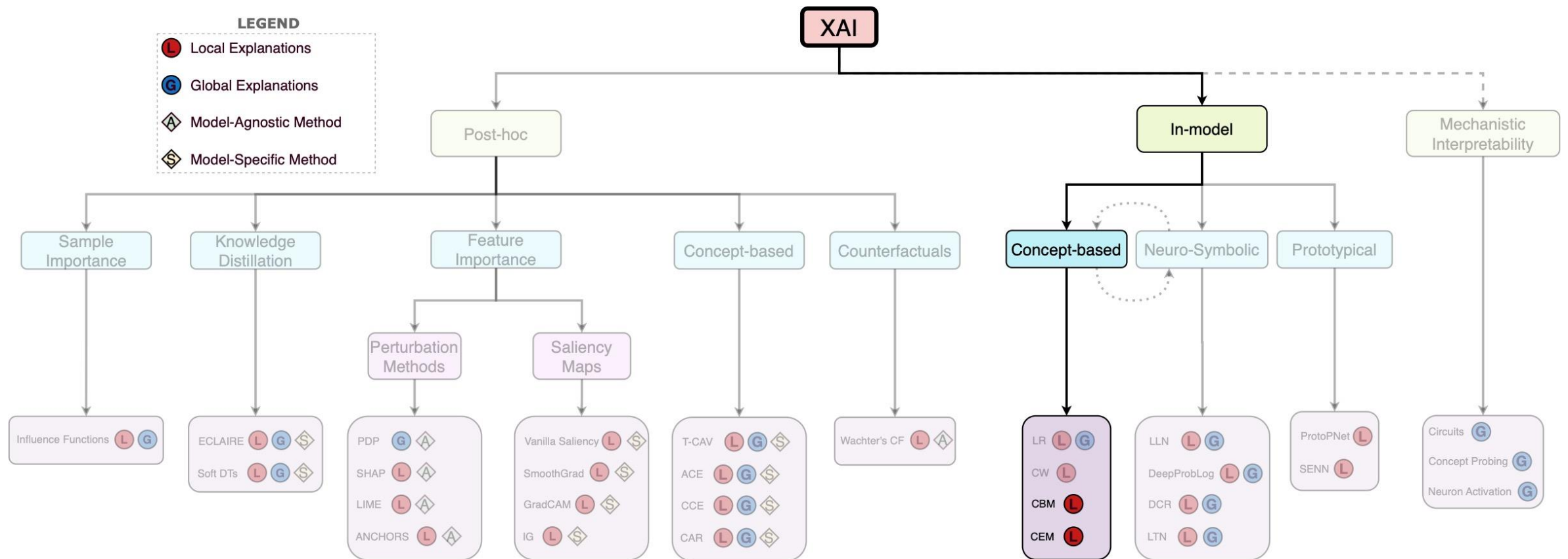
Maybe... but they still have some clear open questions:

What can we do about it?

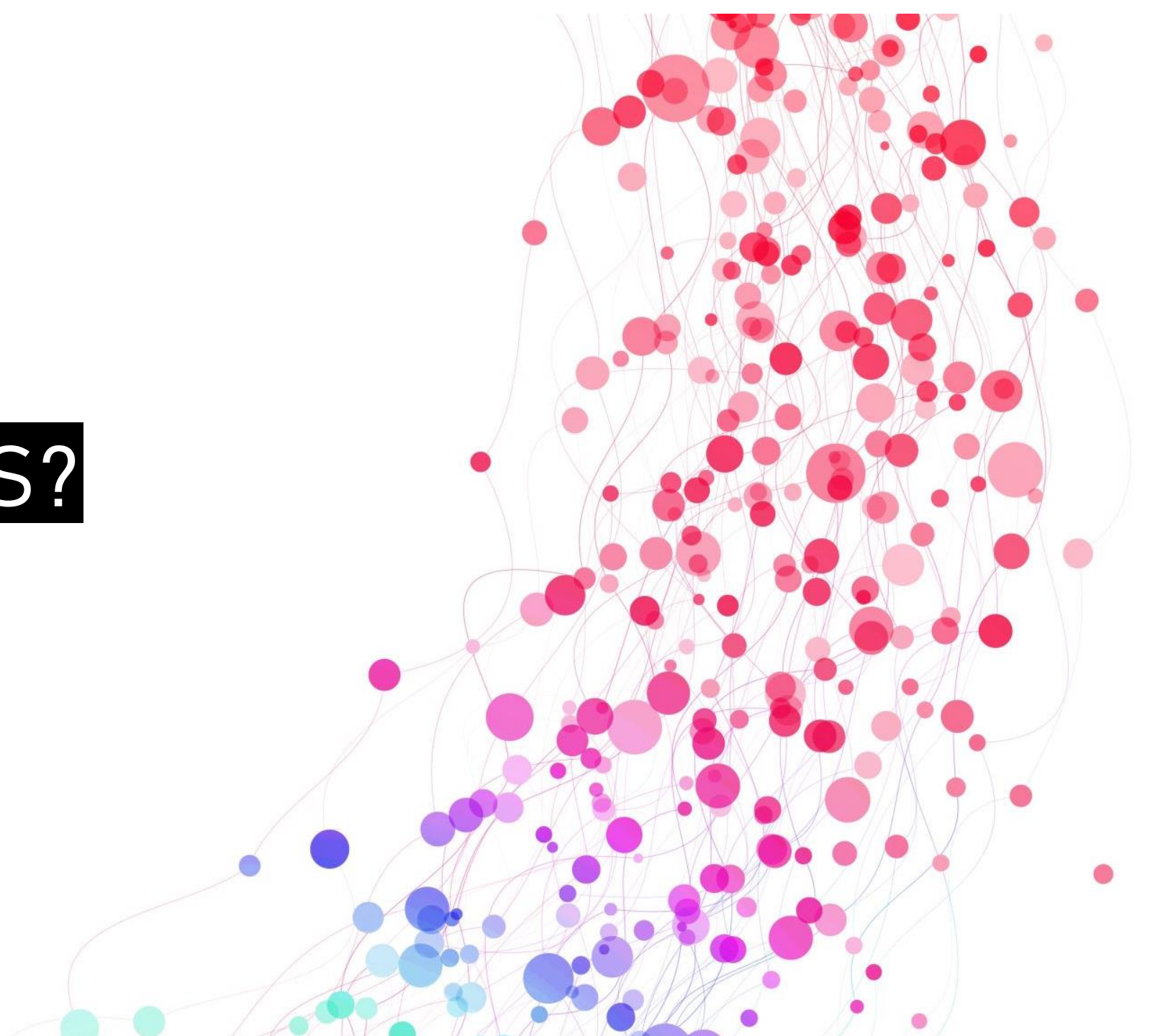


More on this in two weeks!

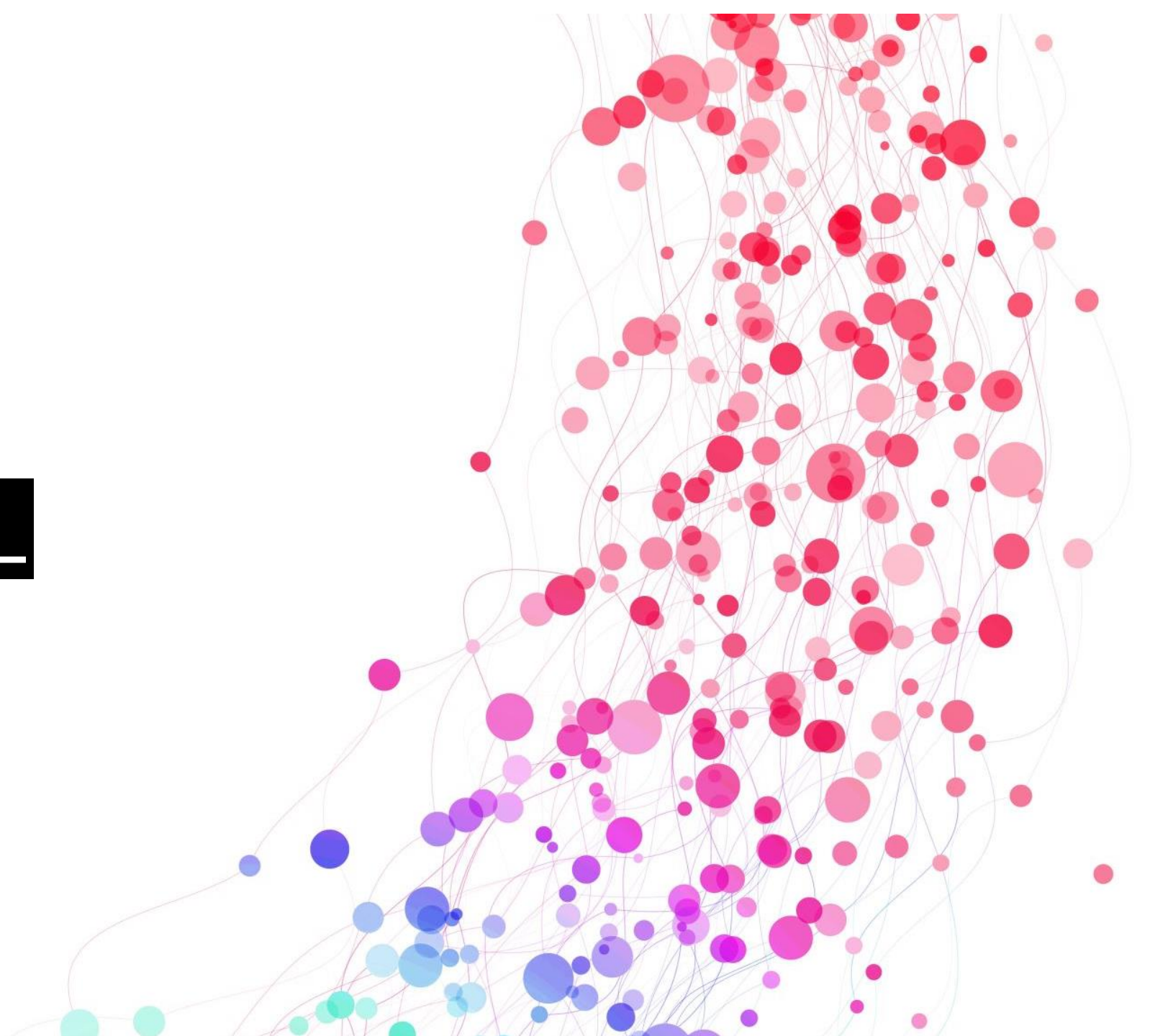
RECAP FOR TODAY



QUESTIONS?



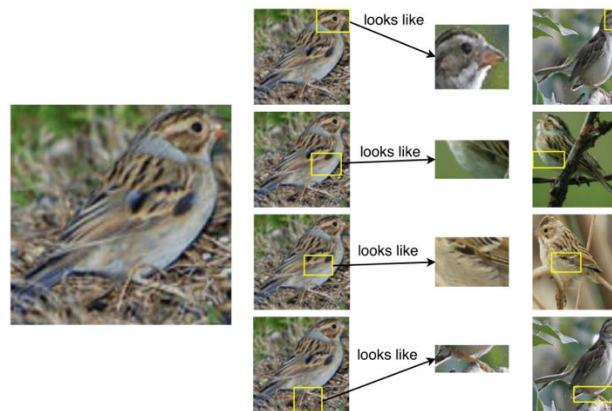
EXTRA MATERIAL



PROTOPNET

CHEN ET AL. (NEURIPS 2019)

Humans very often tend to explain themselves by pointing back at examples from their past experience: "This looks like that..."



ProtoPNet [1]:
"Wouldn't it be nice if we could learn these prototypes in a differentiable manner?"

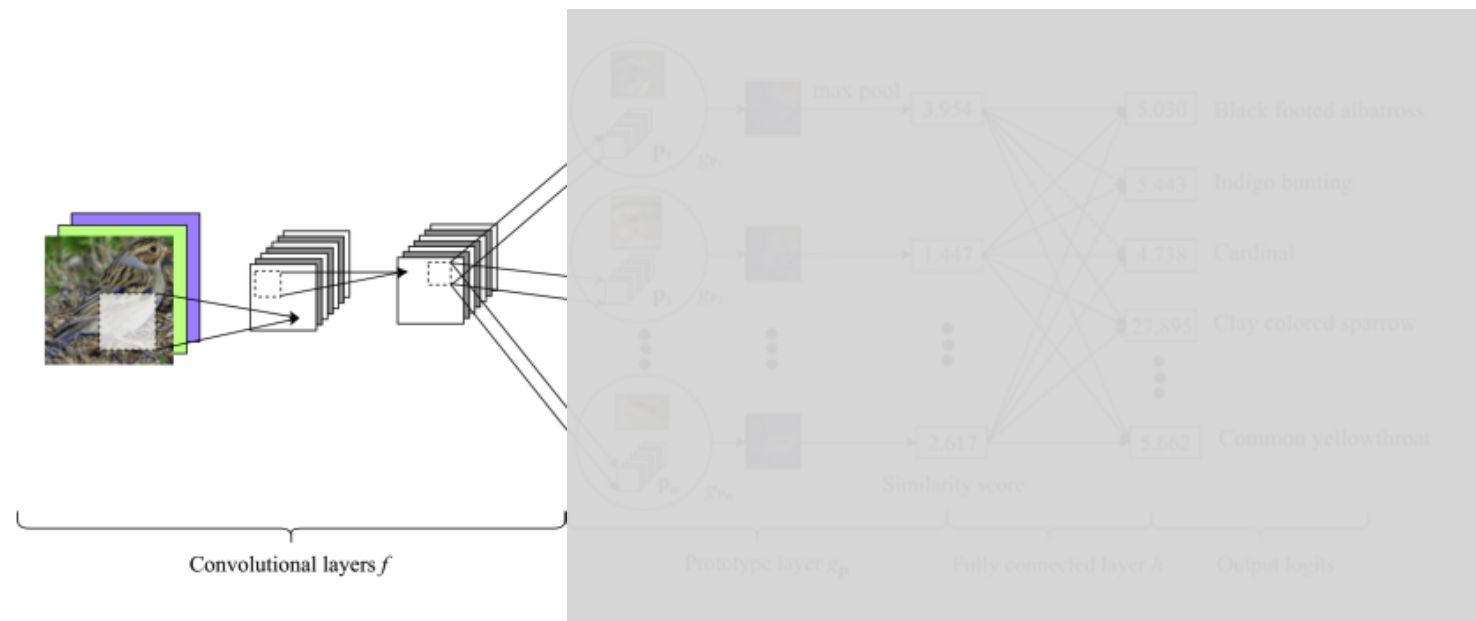
We call this **prototypical explanations** and we construct them by looking at how training samples can be used to explain new predictions



LEARNING PROTOTYPICAL EXPLANATIONS

ProtoPNet divides a neural architecture into **four components**:

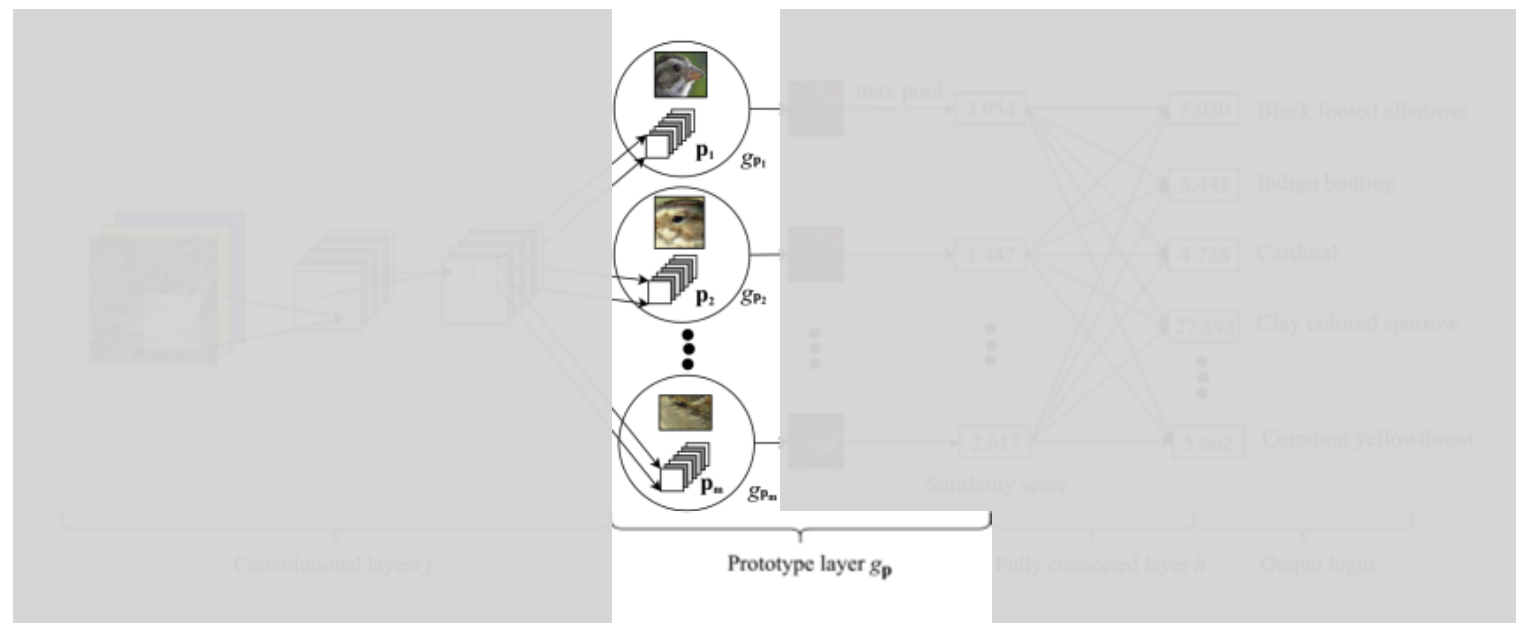
1. We extract 3D representations of our input images through a pretrained model



LEARNING PROTOTYPICAL EXPLANATIONS

ProtoPNet divides a neural architecture into **four components**:

2. Then, we will **learn a prototype layer which holds m prototypical “patches”**.

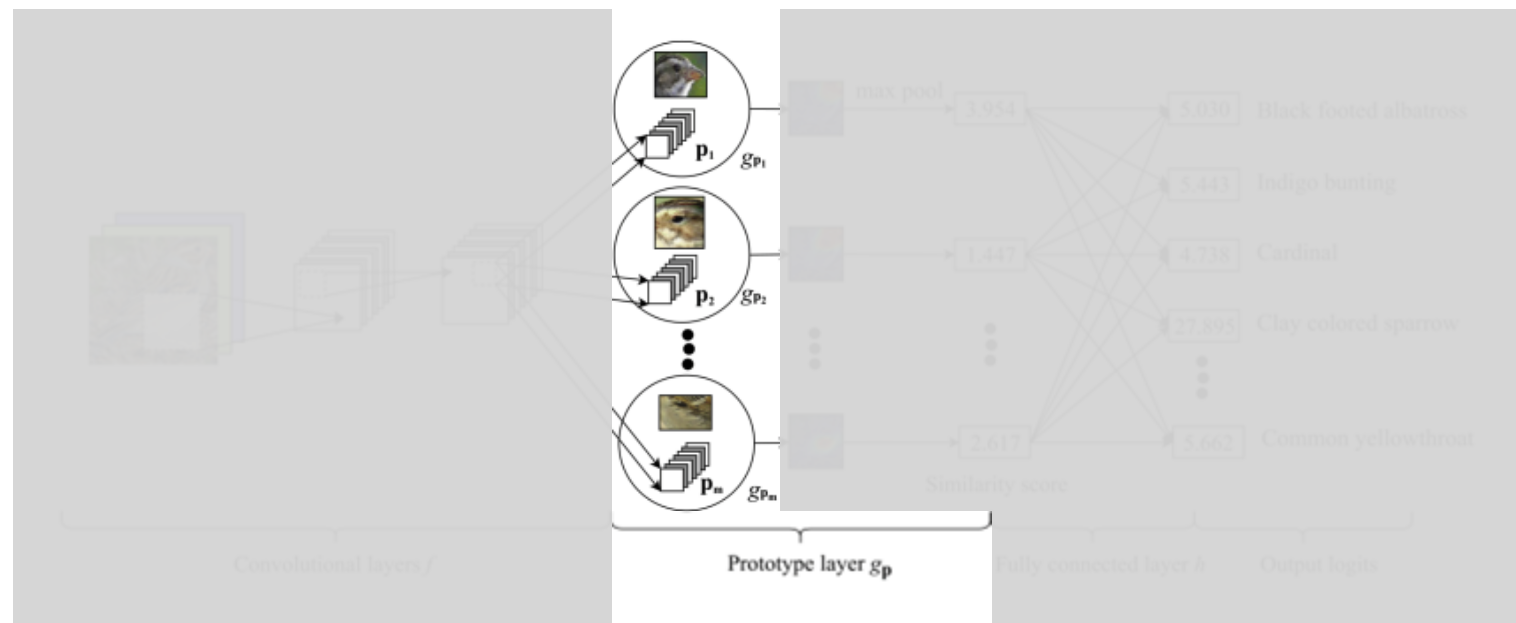


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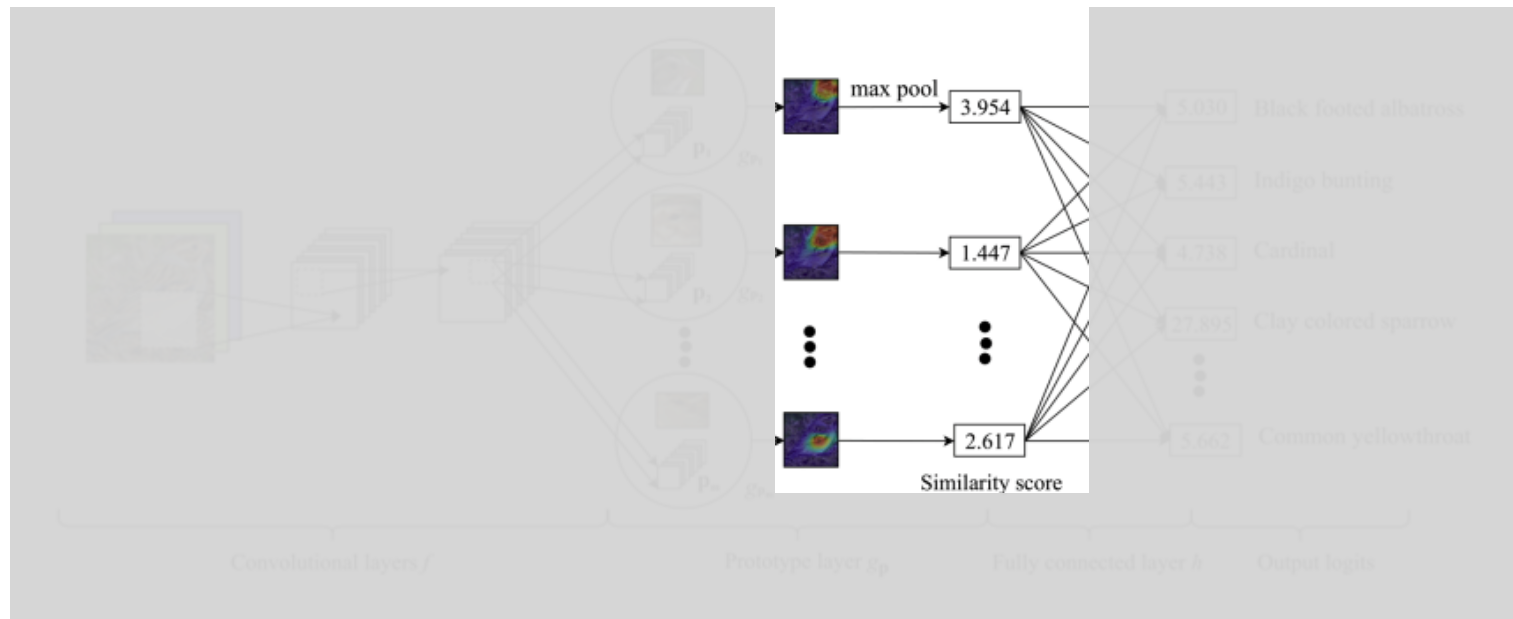
We partition the image's latent code into small patches of the same size as our prototypes!



LEARNING PROTOTYPICAL EXPLANATIONS

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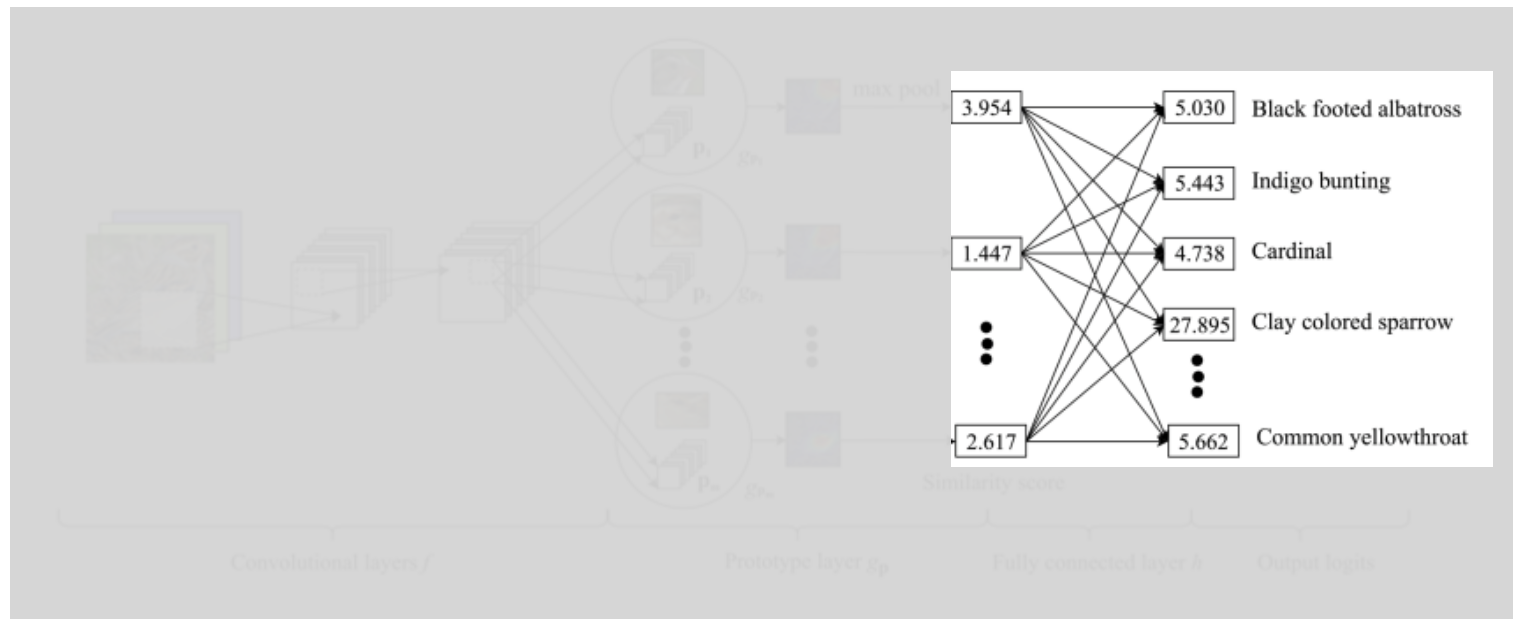
3. We **compute similarities between patches in the input image and our set of learnable prototypes** to get similarities between image patches and prototypes:



LEARNING PROTOTYPICAL EXPLANATIONS

ProtoPNet divides a neural architecture into **four components**:

4. Finally, we output **a prediction** from the set of similarity scores using **an interpretable classifier**



HOW TO TRAIN YOUR PROTOPNET?

We train this architecture by iterating over:

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$$\text{Clst} = \frac{1}{n} \sum_{i=1}^n \min_{j: \mathbf{p}_j \in \mathbf{P}_{y_i}} \min_{\mathbf{z} \in \text{patches}(f(\mathbf{x}_i))} \|\mathbf{z} - \mathbf{p}_j\|_2^2$$

Minimise patch distance for at least one patch in the same class

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 - iii. **Separated** (patches are far from prototypes from other classes)

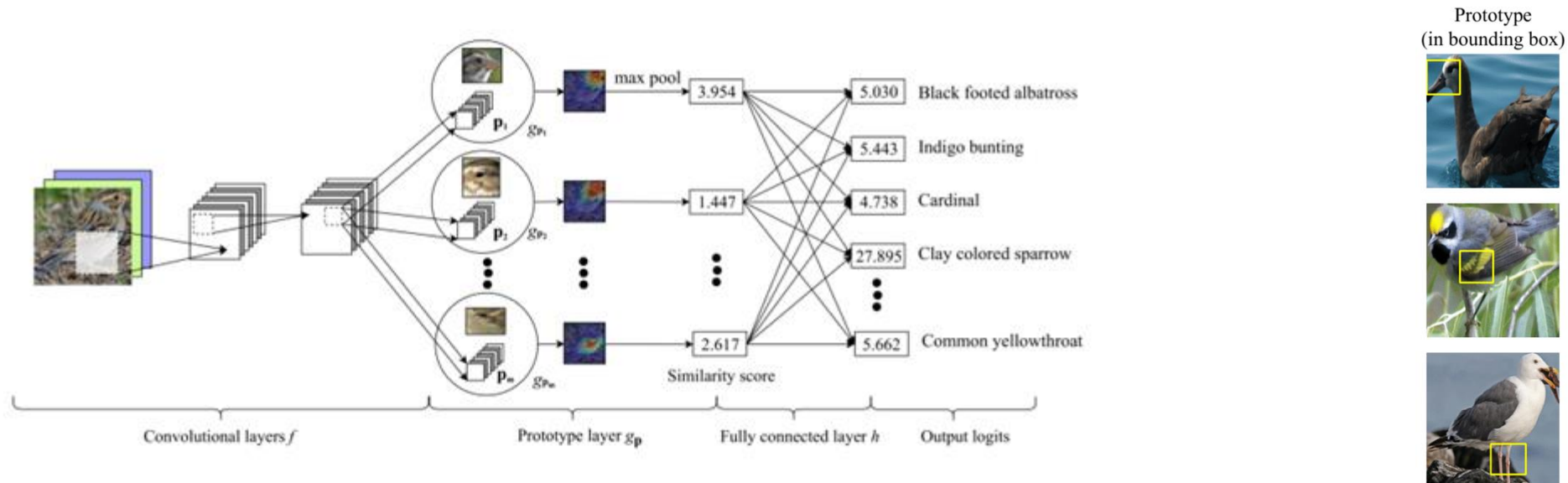
$$\text{Sep} = -\frac{1}{n} \sum_{i=1}^n \min_{j: \mathbf{p}_j \notin \mathbf{P}_{y_i}} \min_{\mathbf{z} \in \text{patches}(f(\mathbf{x}_i))} \|\mathbf{z} - \mathbf{p}_j\|_2^2$$

Maximise patch distance for all patches in other classes

HOW TO TRAIN YOUR PROTOPNET?

We train this architecture by iterating over:

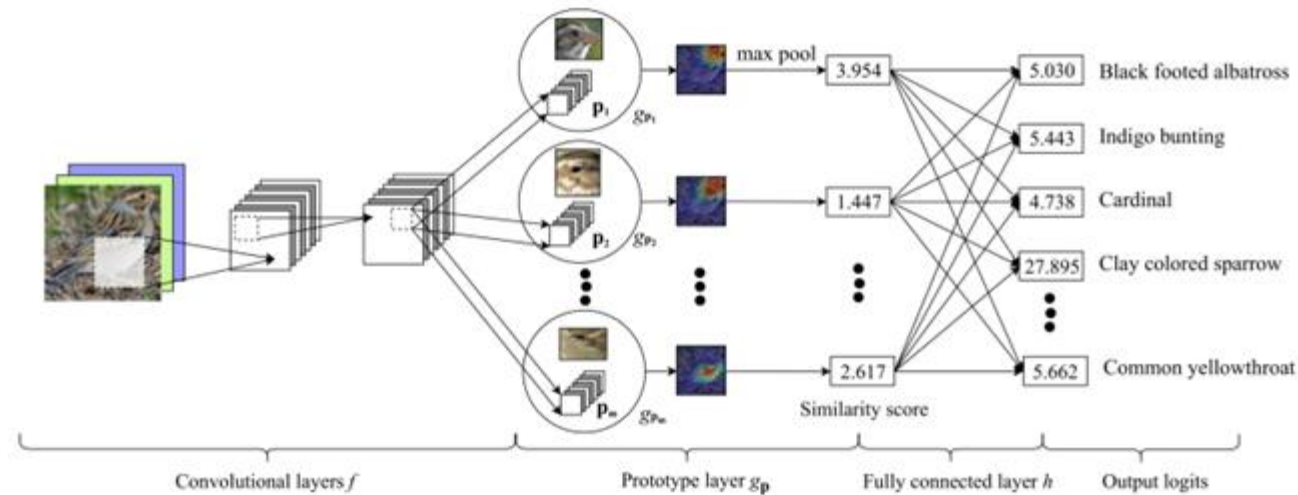
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- Projecting learnt prototypes** to their closest patch in the training set.



HOW TO TRAIN YOUR PROTOPNET?

We train this architecture by iterating over:

- Training the prototype layer.
- Projecting learnt prototypes** to their closest patch in the training set.
- Fine-tuning the output fully connected layer** to map prototype scores to labels.

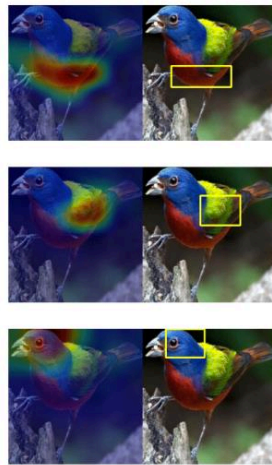


EXPLAINING BY EXAMPLE

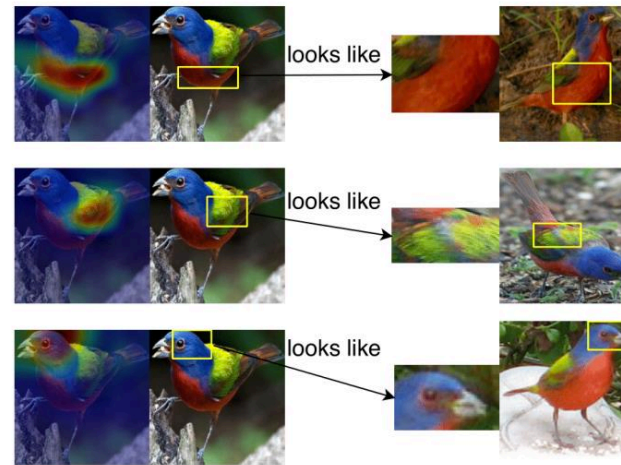
ProtoPNet explains its predictions with **prototypes that are highlighted in the input image with heatmaps** → More interpretable than feature attribution!



(a) Object attention
(class activation map)



(b) Part attention
(attention-based models)



(c) Part attention + comparison with learned
prototypical parts (our model)

ON PROTOTYPICAL EXPLANATIONS

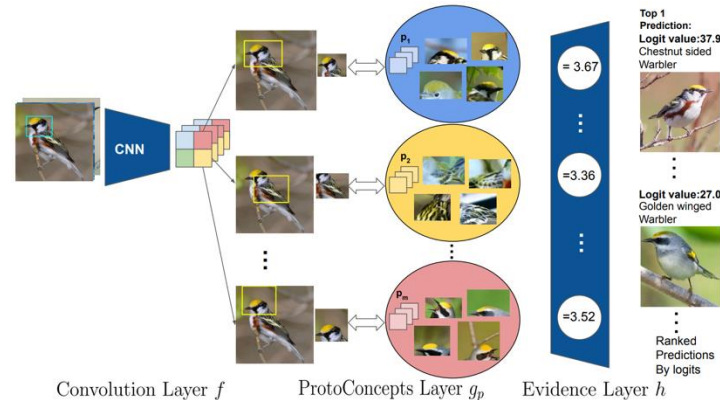
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ON PROTOTYPICAL EXPLANATIONS

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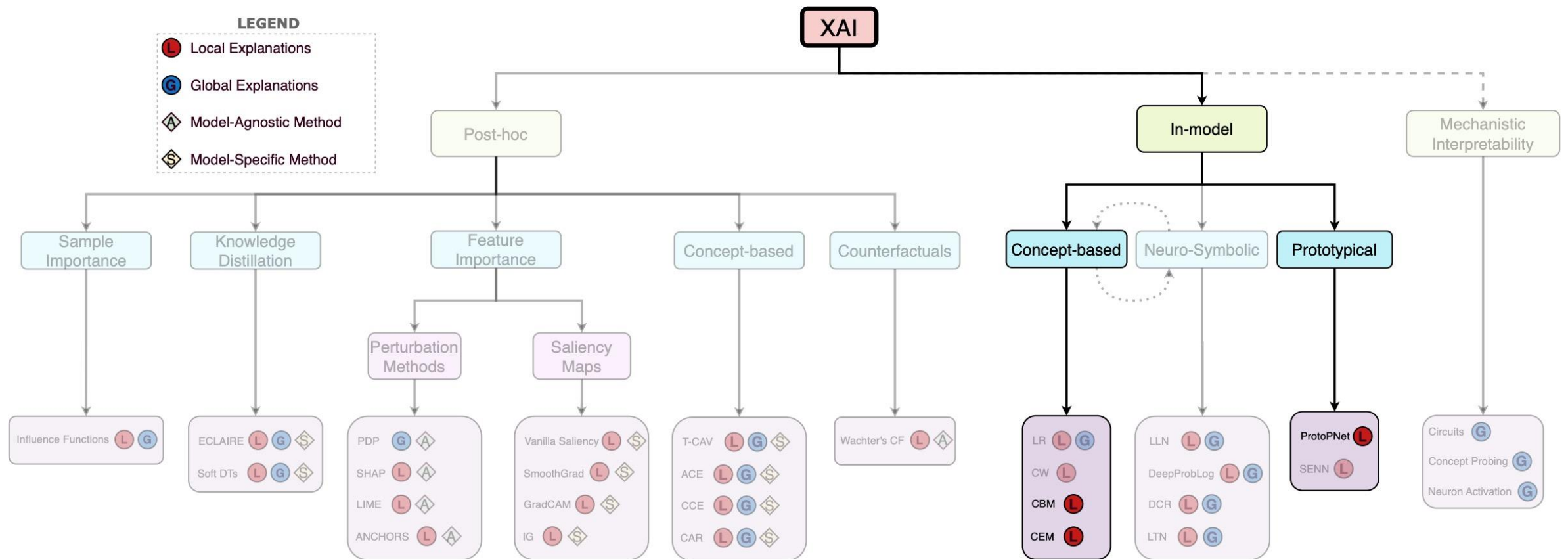


ON PROTOTYPICAL EXPLANATIONS

Prototypes are certainly an interesting research direction! However:

1. They are **prone to complicated training pipelines** as it is non-trivial to learn "traceable" prototypes in a differentiable manner
2. They tend to select **only single patches/parts of examples as prototypes**, complicating disambiguating the reason behind selecting that prototype
3. Part prototypes are not very useful in a lot of domains such as **language and genomics** (e.g., when trying to understand memorisation)

RECAP FOR TODAY



QUESTIONS?

